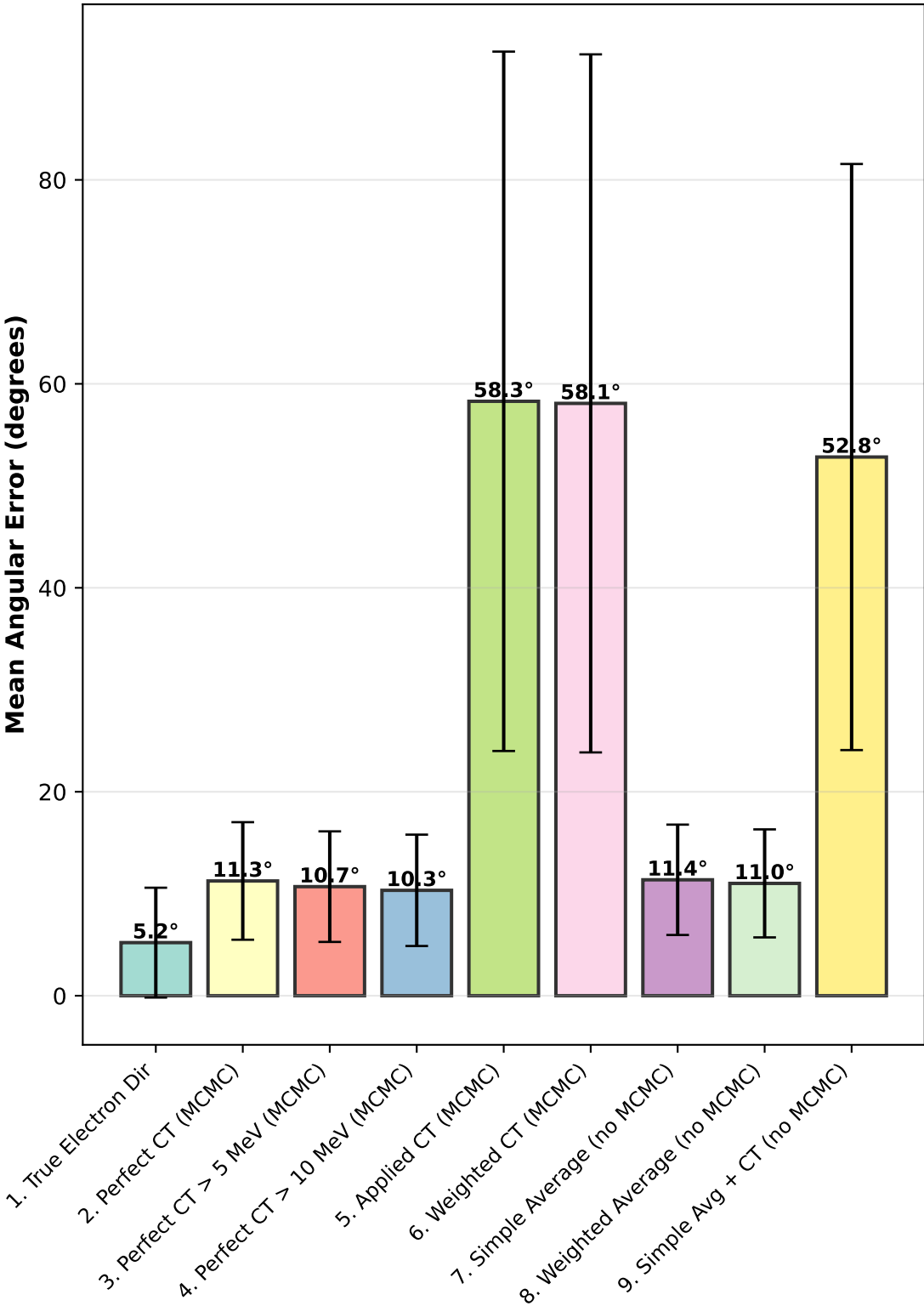


Emcee 8-Scenario Aggregate Analysis
Avg ES clusters: 1.: 271 ES | 2.: 271 ES | 3.: 241 ES | 4.: 147 ES
5.: 3242 ES | 6.: 3242 ES | 7.: 271 ES | 8.: 271 ES | 9.: 3242 ES

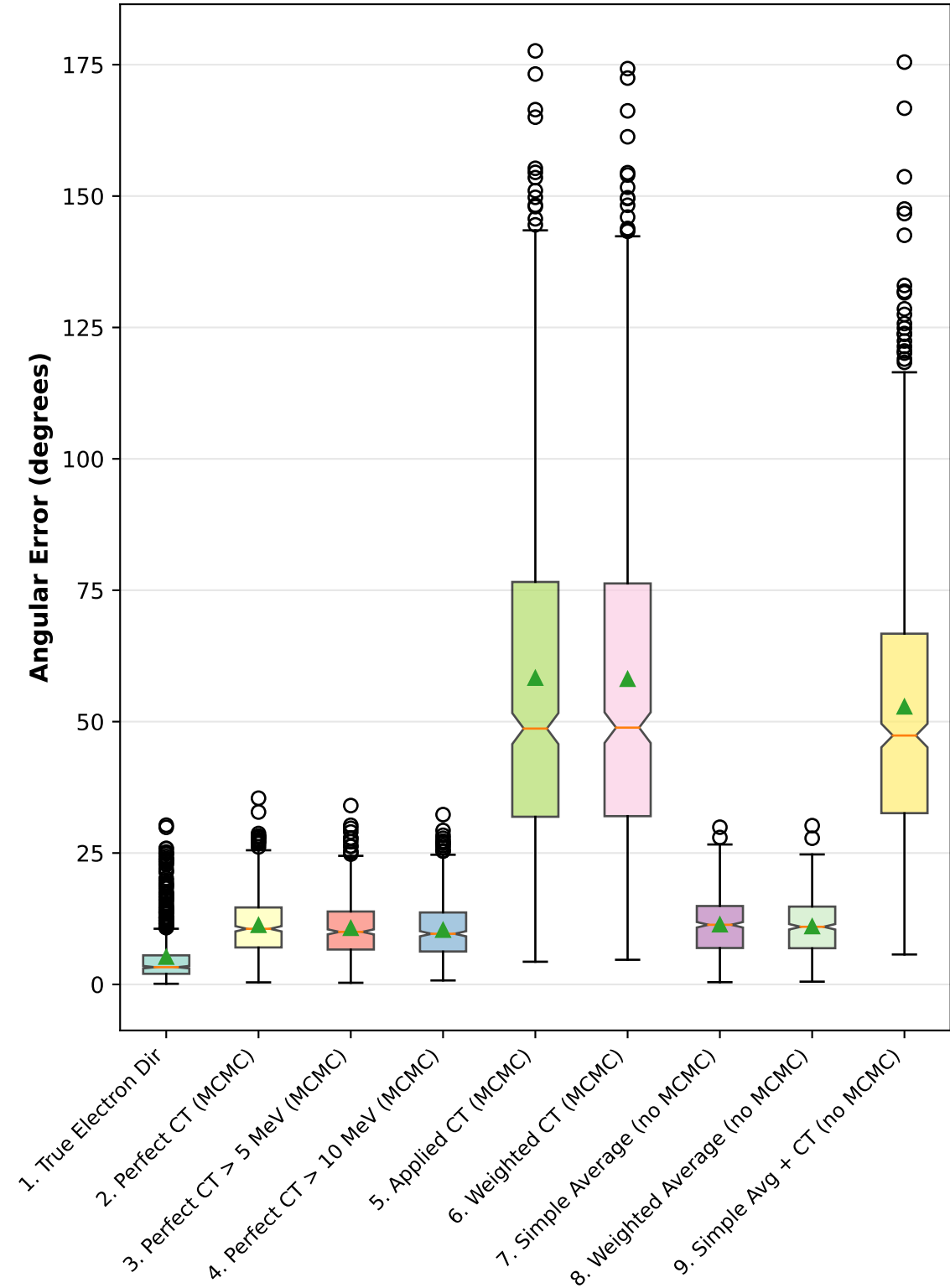
Summary Statistics

Scenario	N	68% Q	Median	Mean
1. True Electron Dir	567	4.72°	3.28°	5.21°
2. Perfect CT (MCMC)	567	13.25°	10.58°	11.26°
3. Perfect CT > 5 MeV (MCMC)	567	12.54°	10.00°	10.70°
4. Perfect CT > 10 MeV (MCMC)	567	12.17°	9.64°	10.34°
5. Applied CT (MCMC)	567	67.16°	48.68°	58.29°
6. Weighted CT (MCMC)	567	67.21°	48.86°	58.08°
7. Simple Average (no MCMC)	567	13.86°	11.35°	11.37°
8. Weighted Average (no MCMC)	567	13.57°	10.96°	11.02°
9. Simple Avg + CT (no MCMC)	567	59.18°	47.36°	52.82°

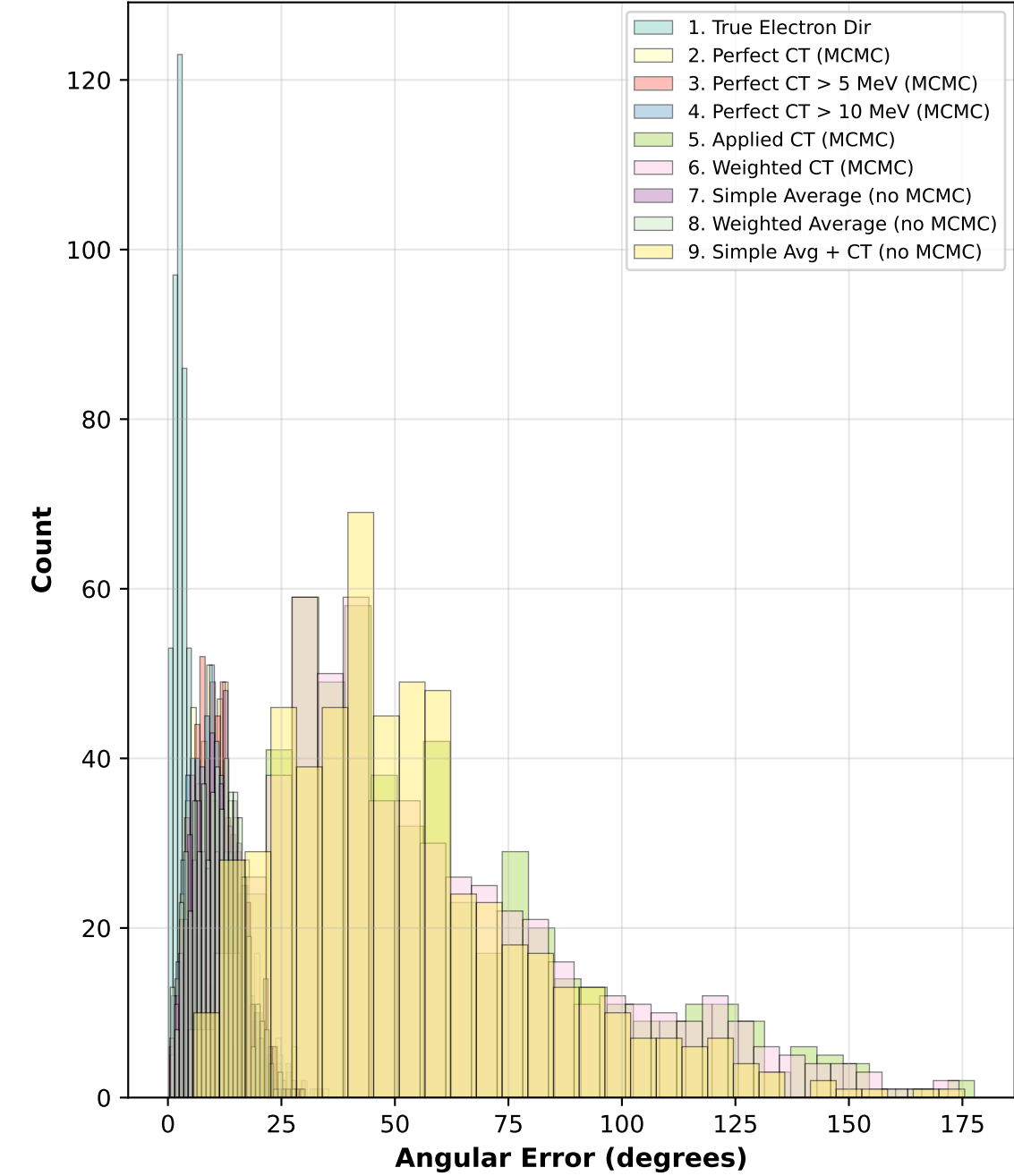
Mean Resolution ± Std Dev



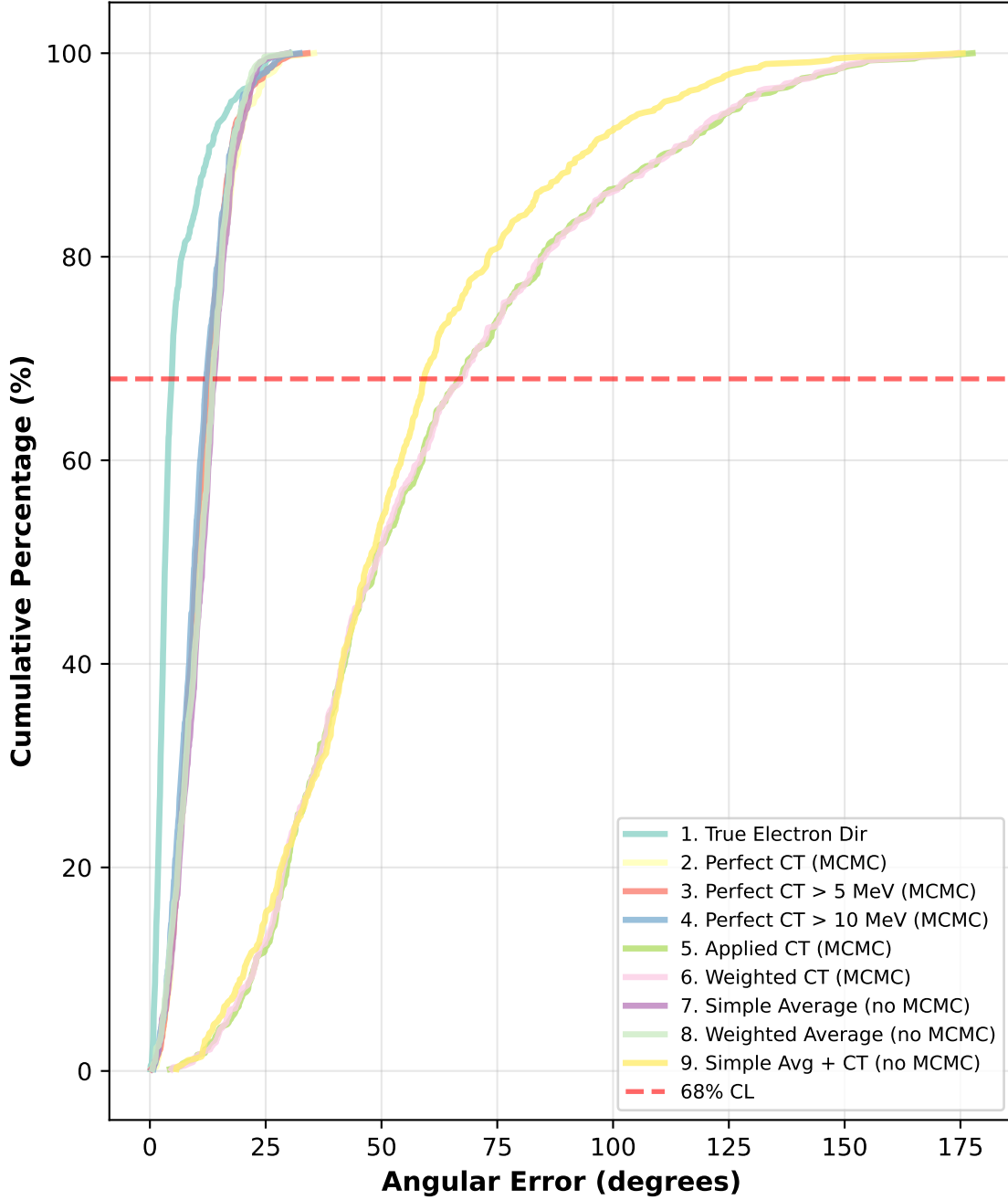
Angular Resolution Comparison



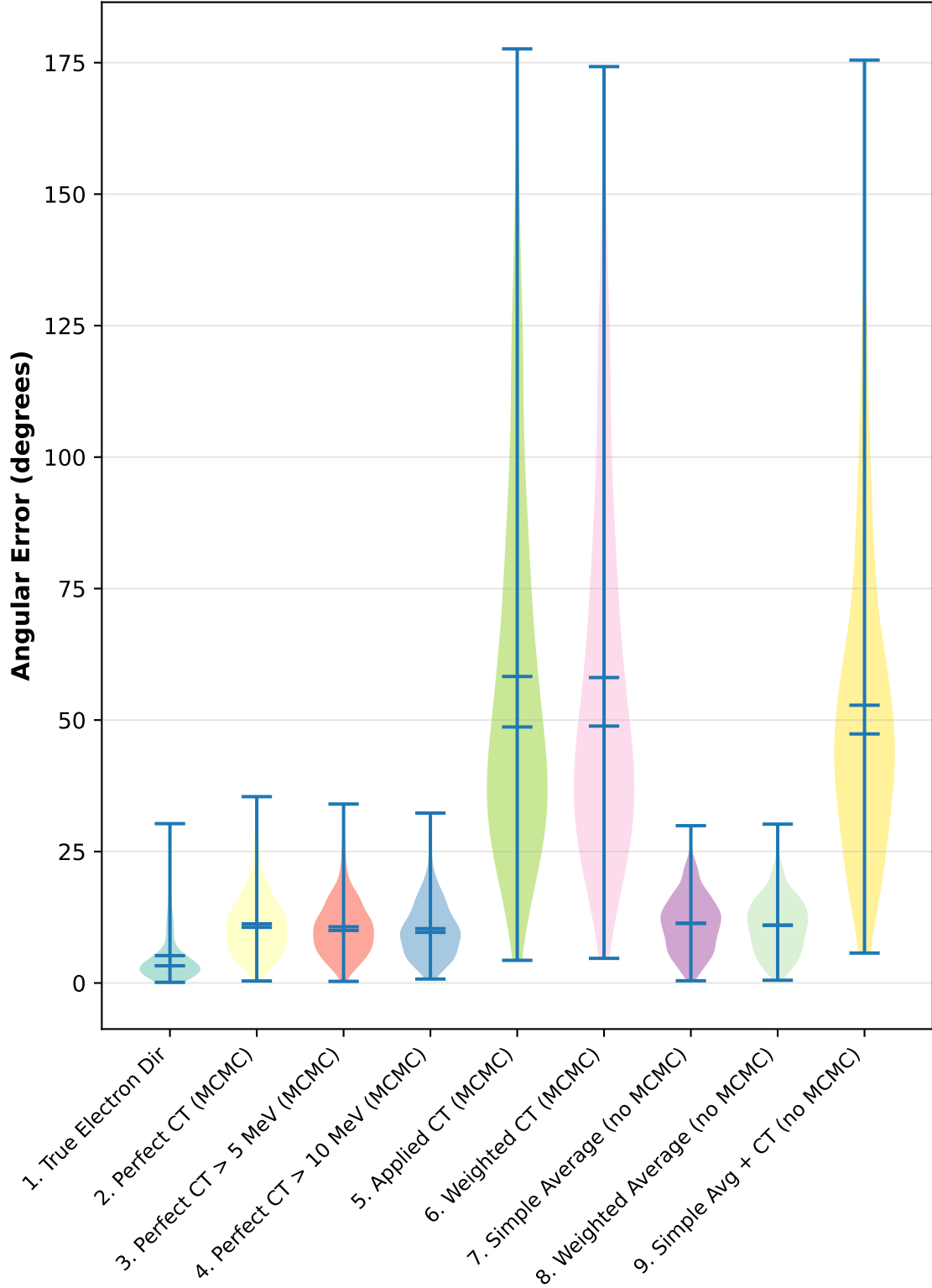
Distribution Overlay



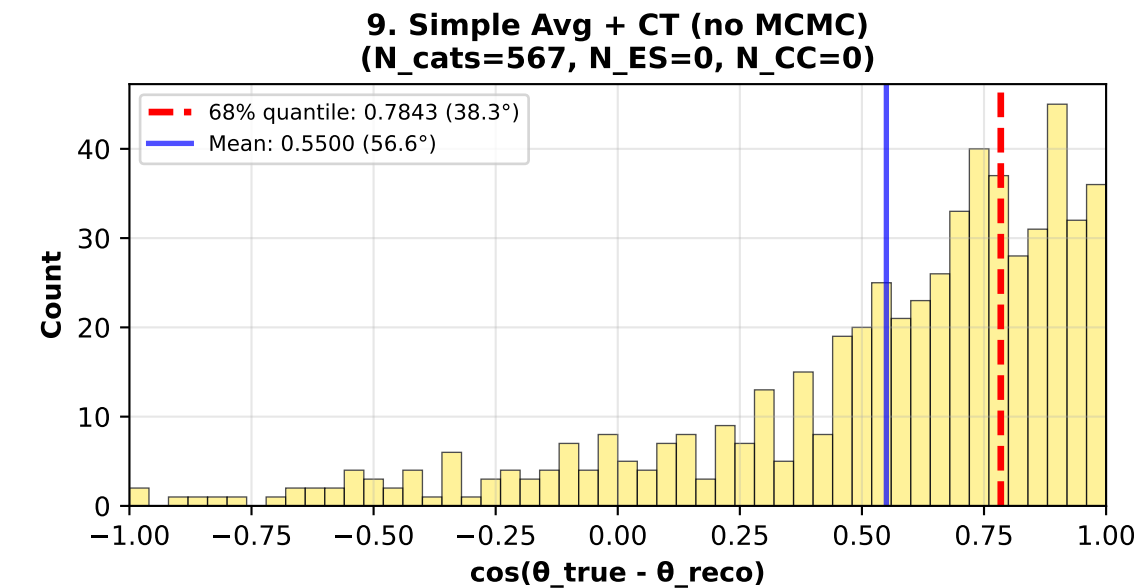
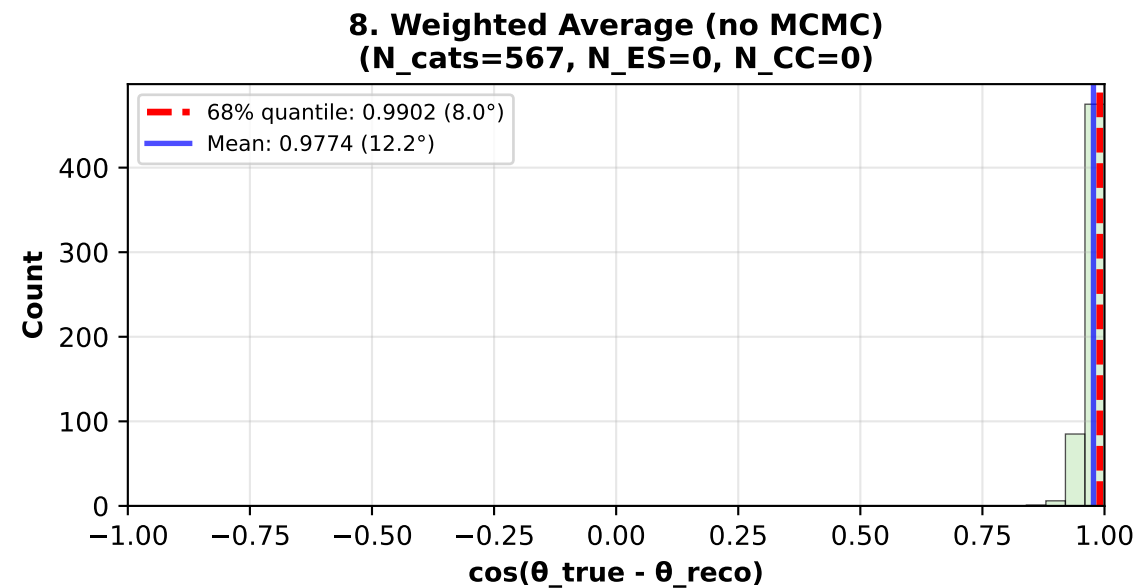
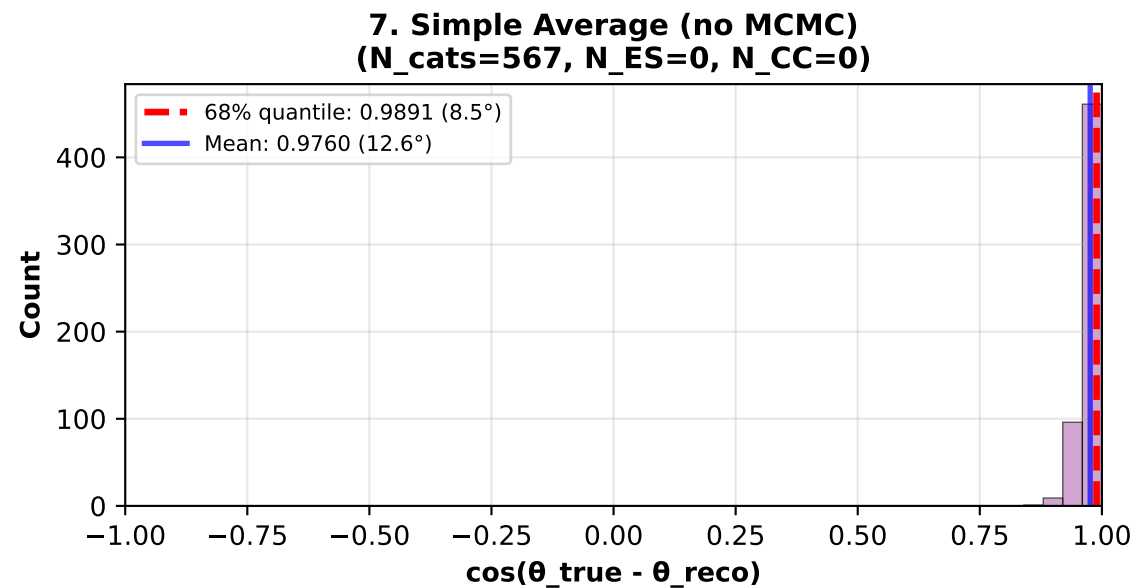
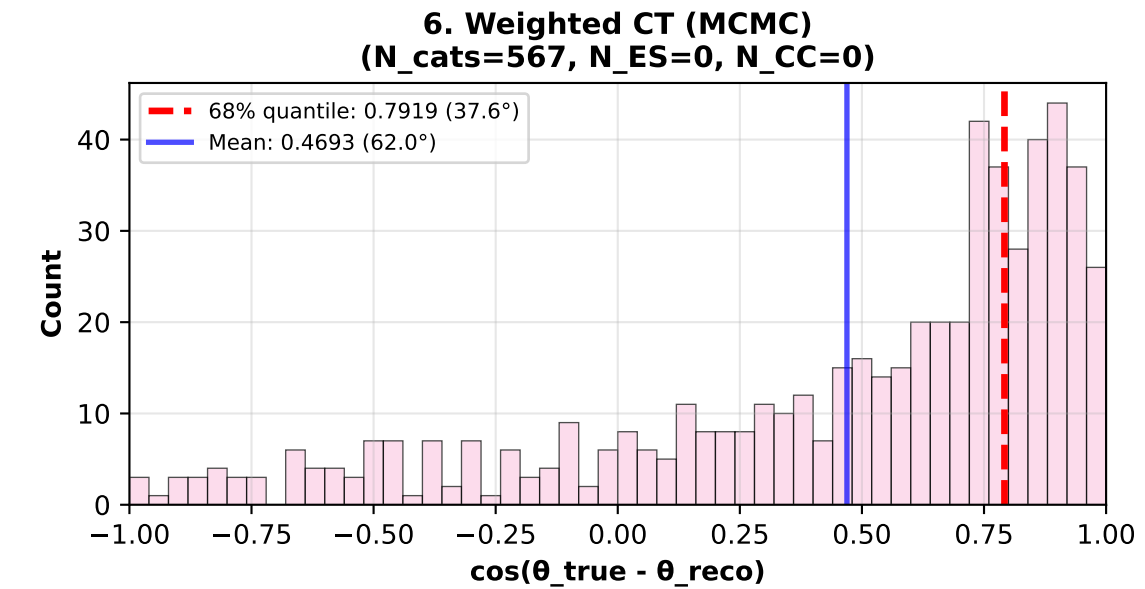
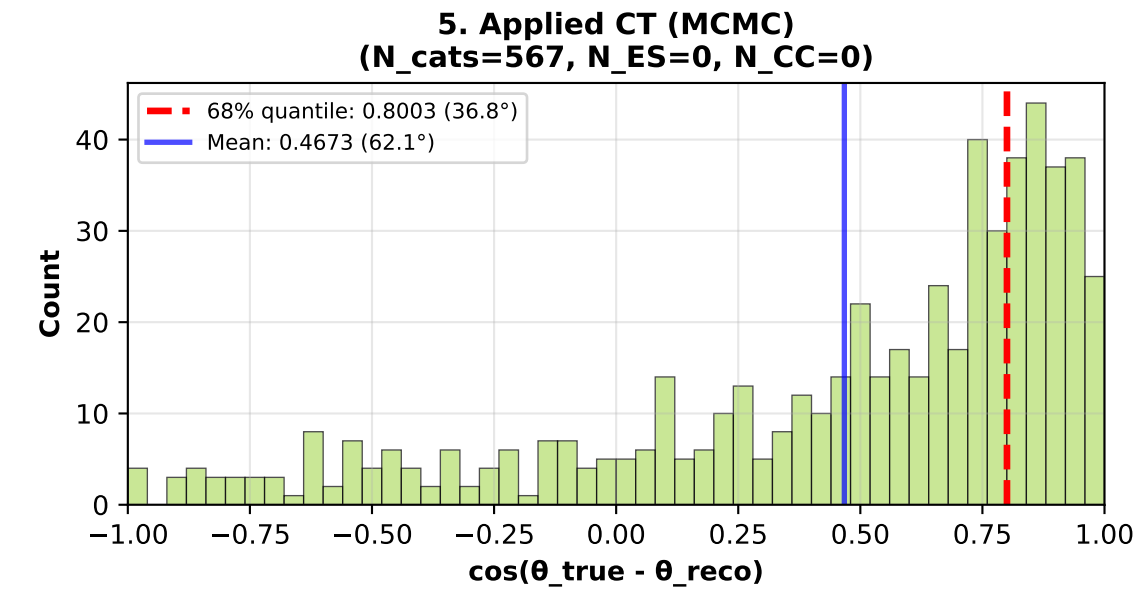
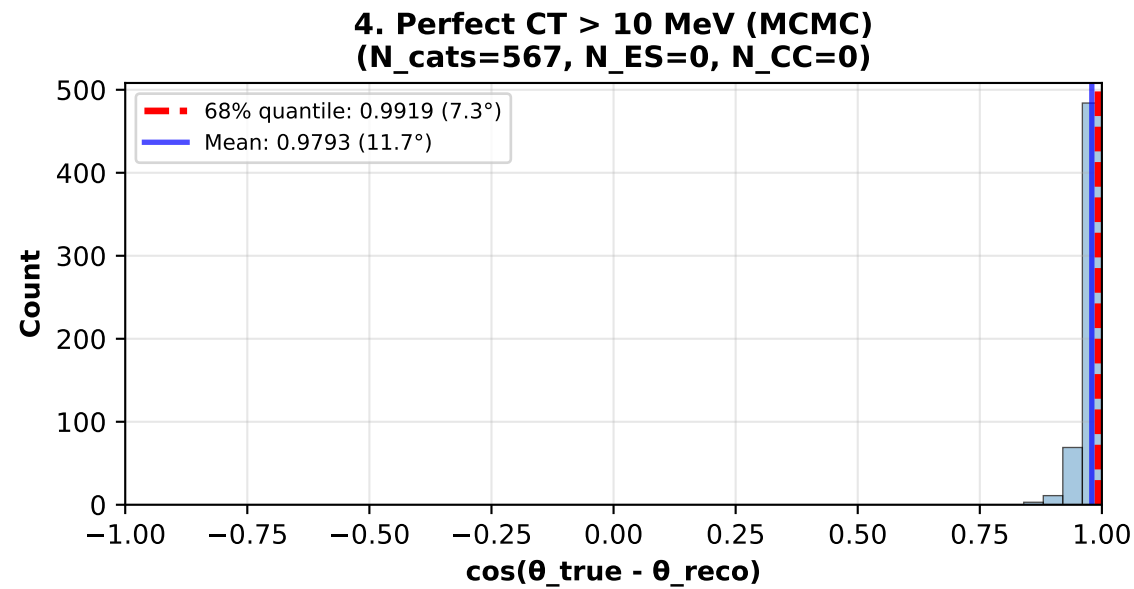
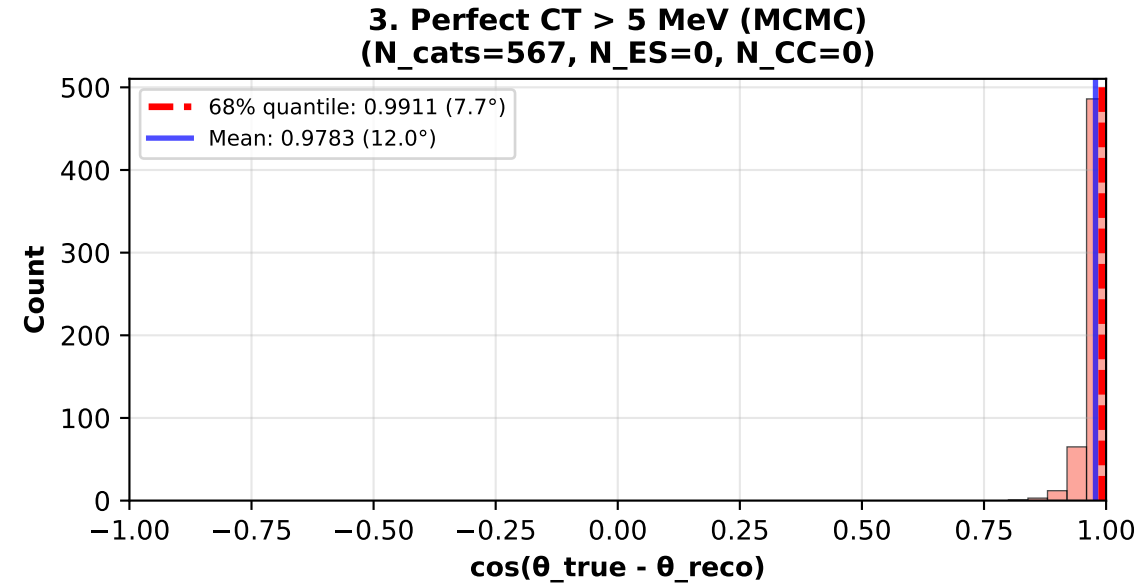
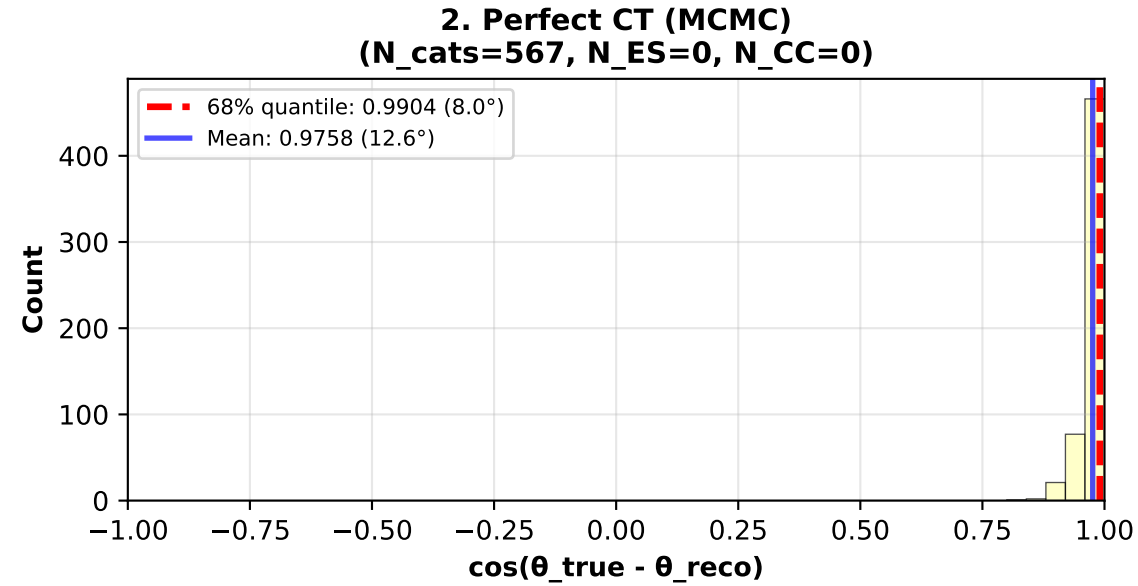
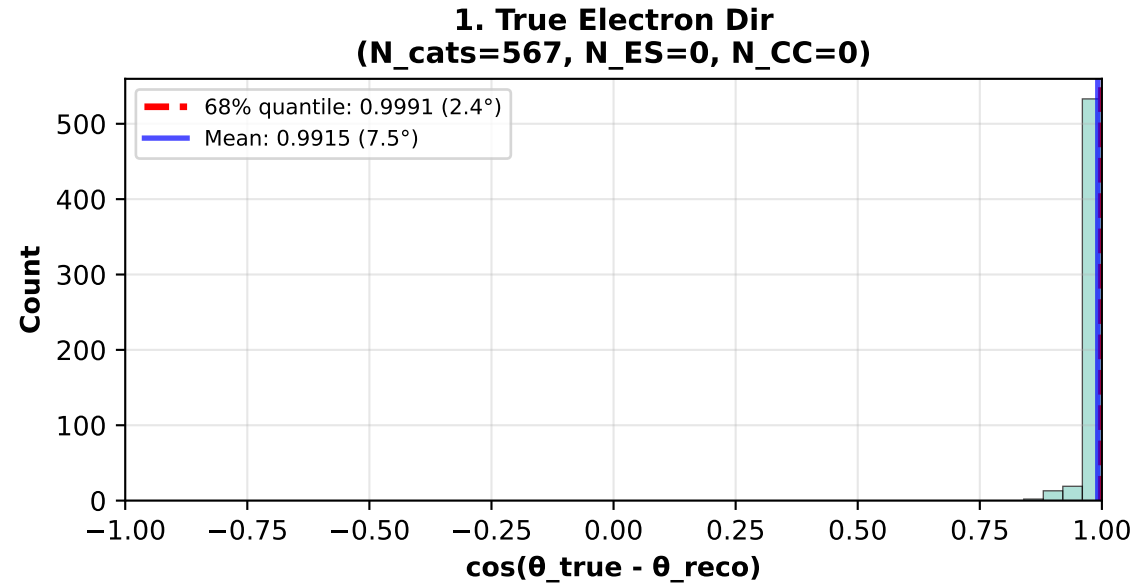
Cumulative Distribution



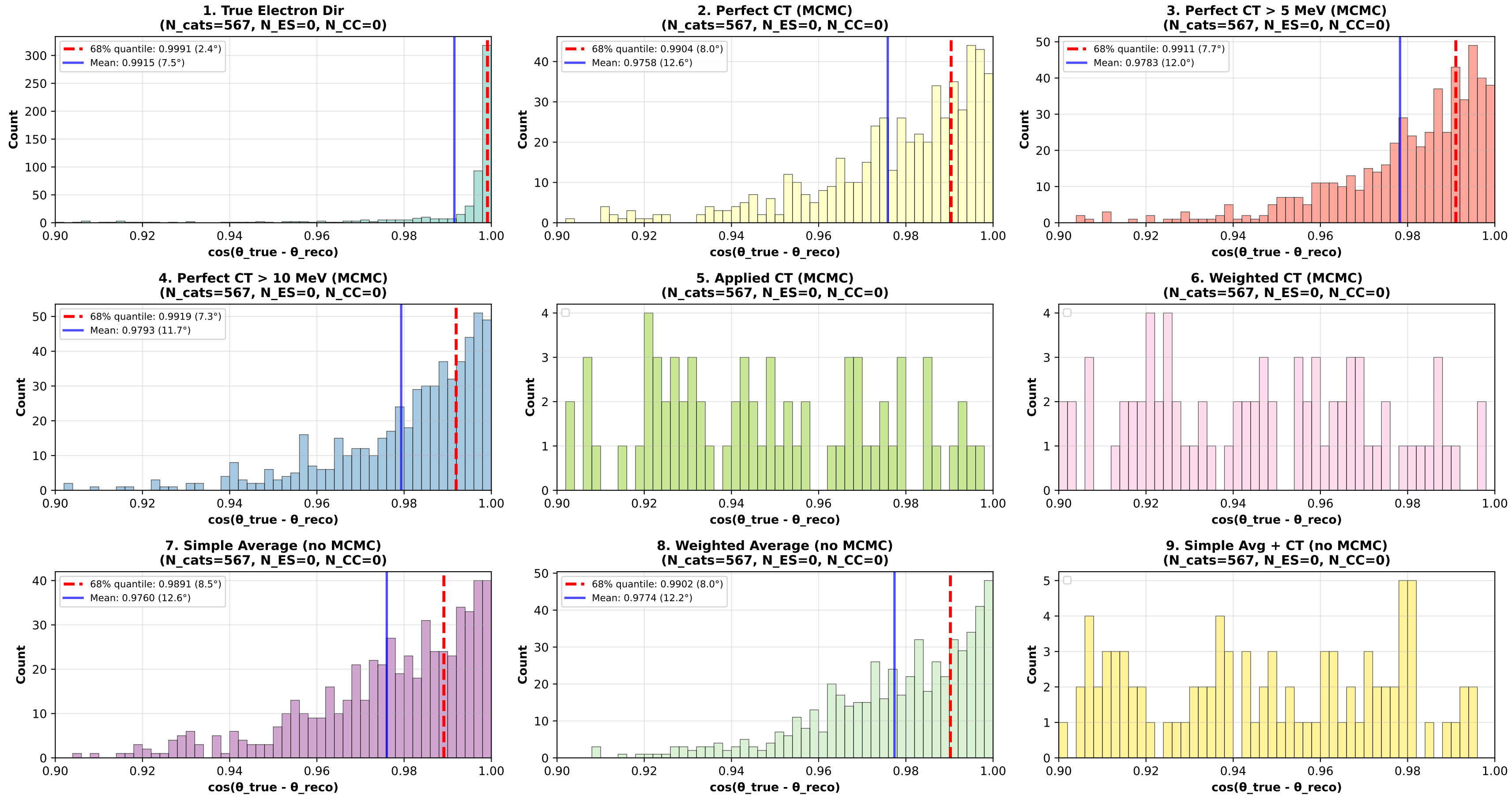
Distribution Shapes



Cosine Distribution: $\cos(\theta_{\text{true}} - \theta_{\text{reco}})$ - Full Range



Cosine Distribution: $\cos(\theta_{\text{true}} - \theta_{\text{reco}})$ - Zoomed (0.9-1.0)



Cluster Usage Across Scenarios

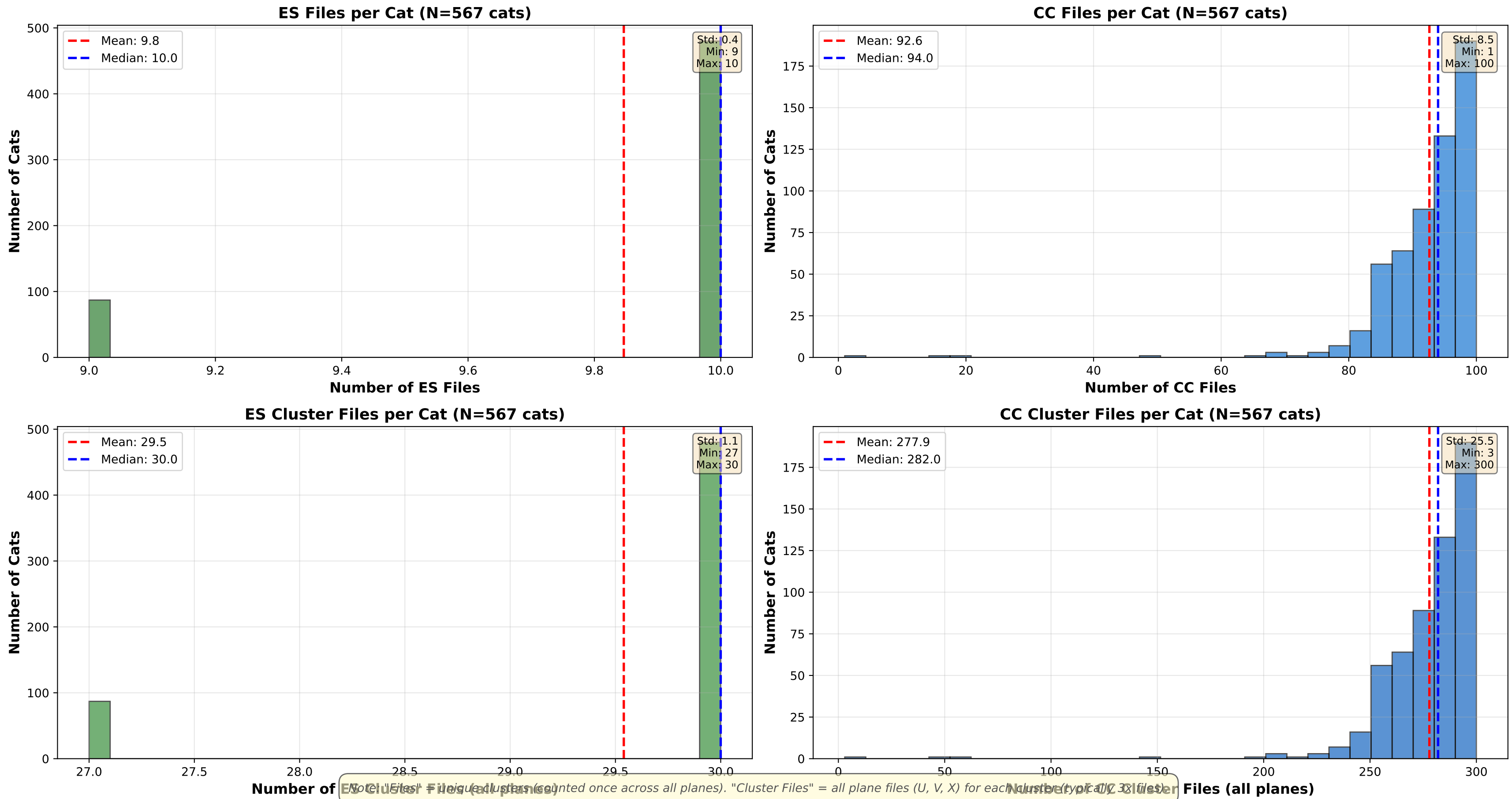
Scenario	N Cats	Input Total	Input ES Main	Value "Clusters Used"	Notes
1. True Electron Dir	567	102	10	271	MCMC samples (weighted sum)
2. Perfect CT (MCMC)	567	102	10	271	MCMC samples (weighted sum)
3. Perfect CT > 5 MeV (MCMC)	567	102	10	241	MCMC samples (weighted sum)
4. Perfect CT > 10 MeV (MCMC)	567	102	10	147	143.6% retention
5. Applied CT (MCMC)	567	102	10	3242	MCMC samples (weighted sum)
6. Weighted CT (MCMC)	567	102	10	3242	MCMC samples (weighted sum)
7. Simple Average (no MCMC)	567	102	10	271	MCMC samples (weighted sum)
8. Weighted Average (no MCMC)	567	102	10	271	MCMC samples (weighted sum)
9. Simple Avg + CT (no MCMC)	567	102	10	3242	MCMC samples (weighted sum)

Note: "Clusters Used" shows different meanings per scenario:

- Best Case, Perfect CT: Number of ES main clusters actually used (after E>3 MeV cut)
- Full Pipeline, Weighted CT: Weighted sum of cluster contributions in MCMC (includes CT probabilities)
- Energy cut scenarios: Number of clusters passing energy threshold

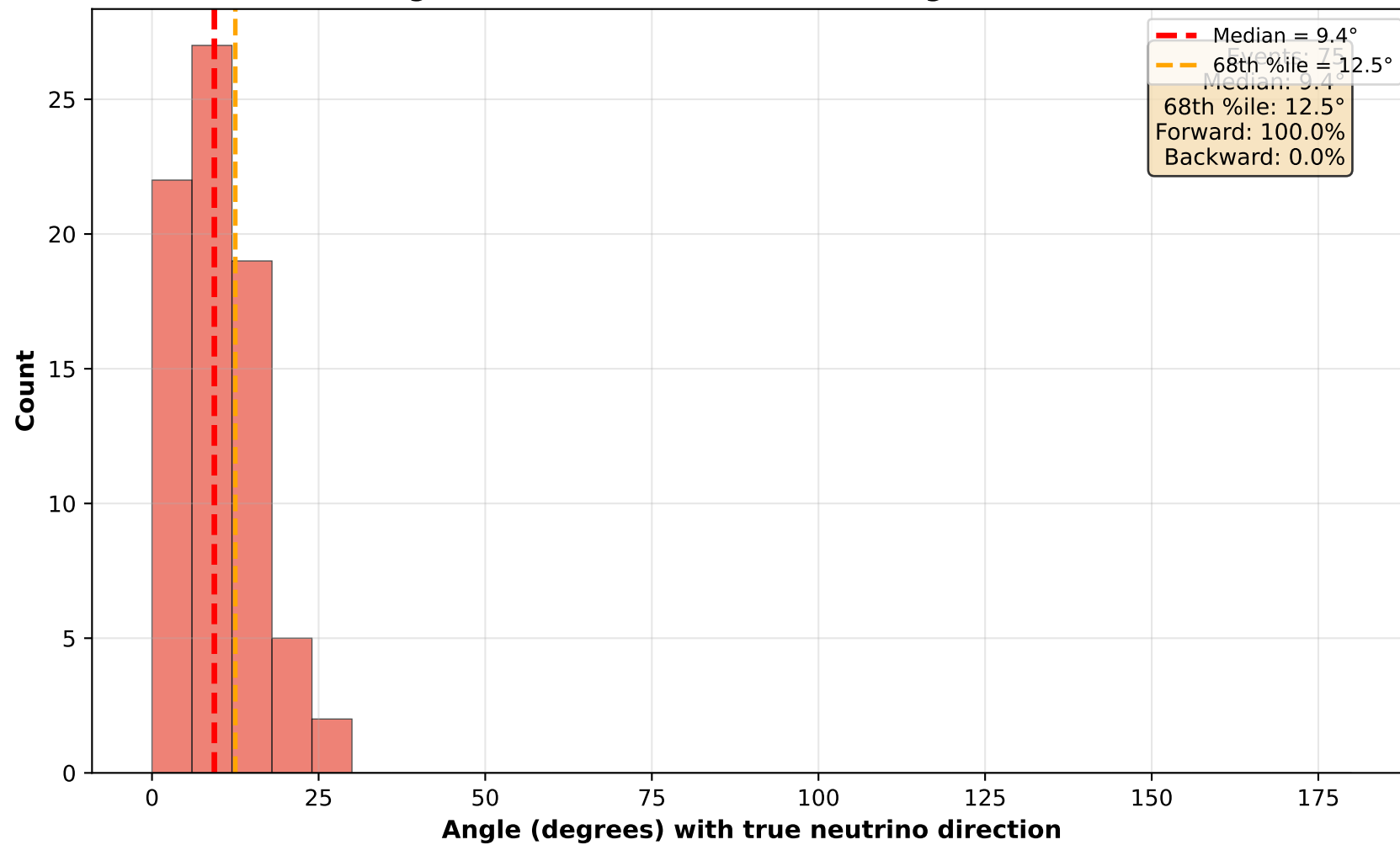
The key insight: Best Case and Perfect CT use ~85-90% of ES main tracks with E>3 MeV.
Full Pipeline shows much higher values due to weighted MCMC sampling, not actual cluster count.

Input Data Statistics: Files and Clusters per Cat

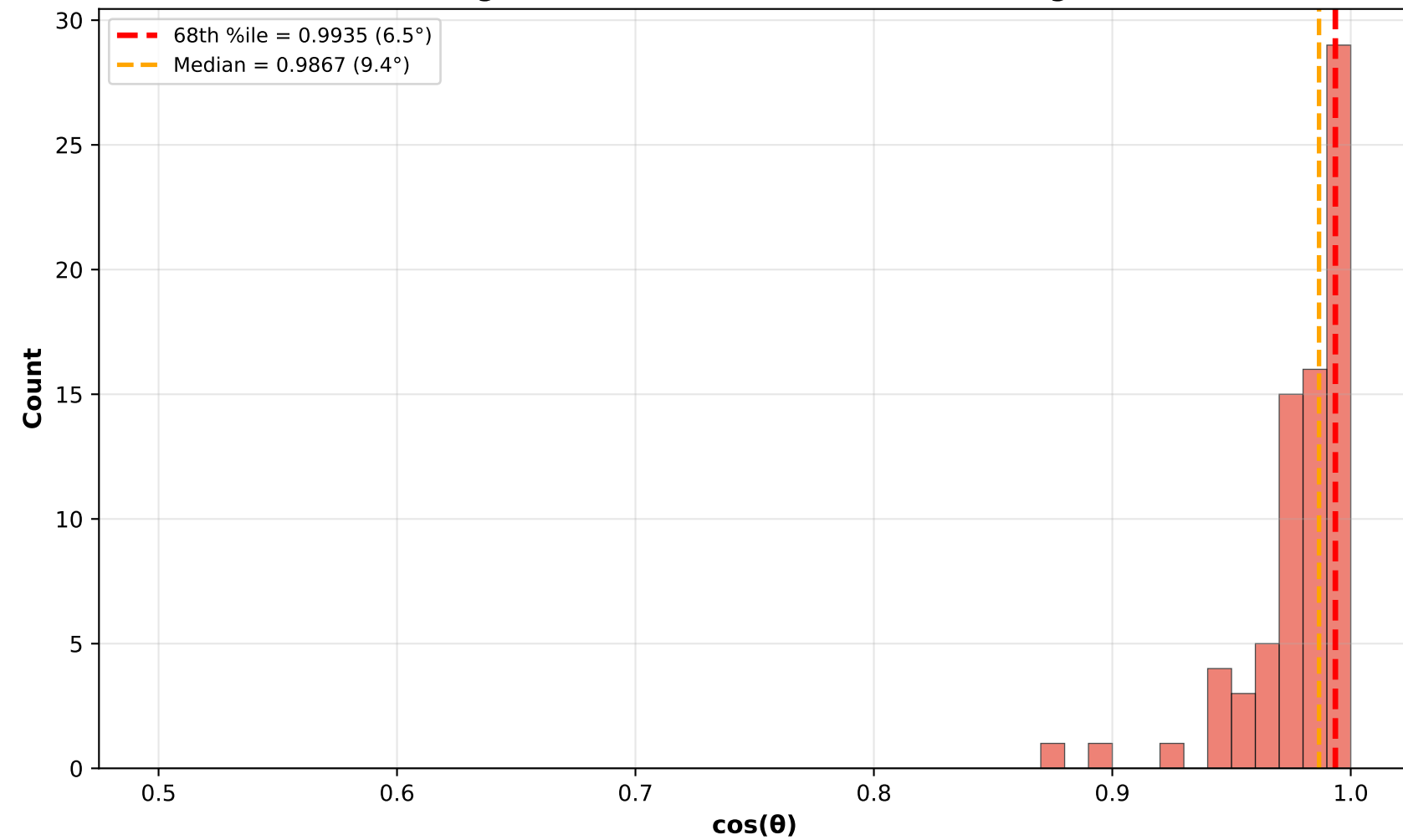


Perfect CT (ES + ED) - Axis-Aligned Analysis ($\pm 30.0^\circ$ selection)

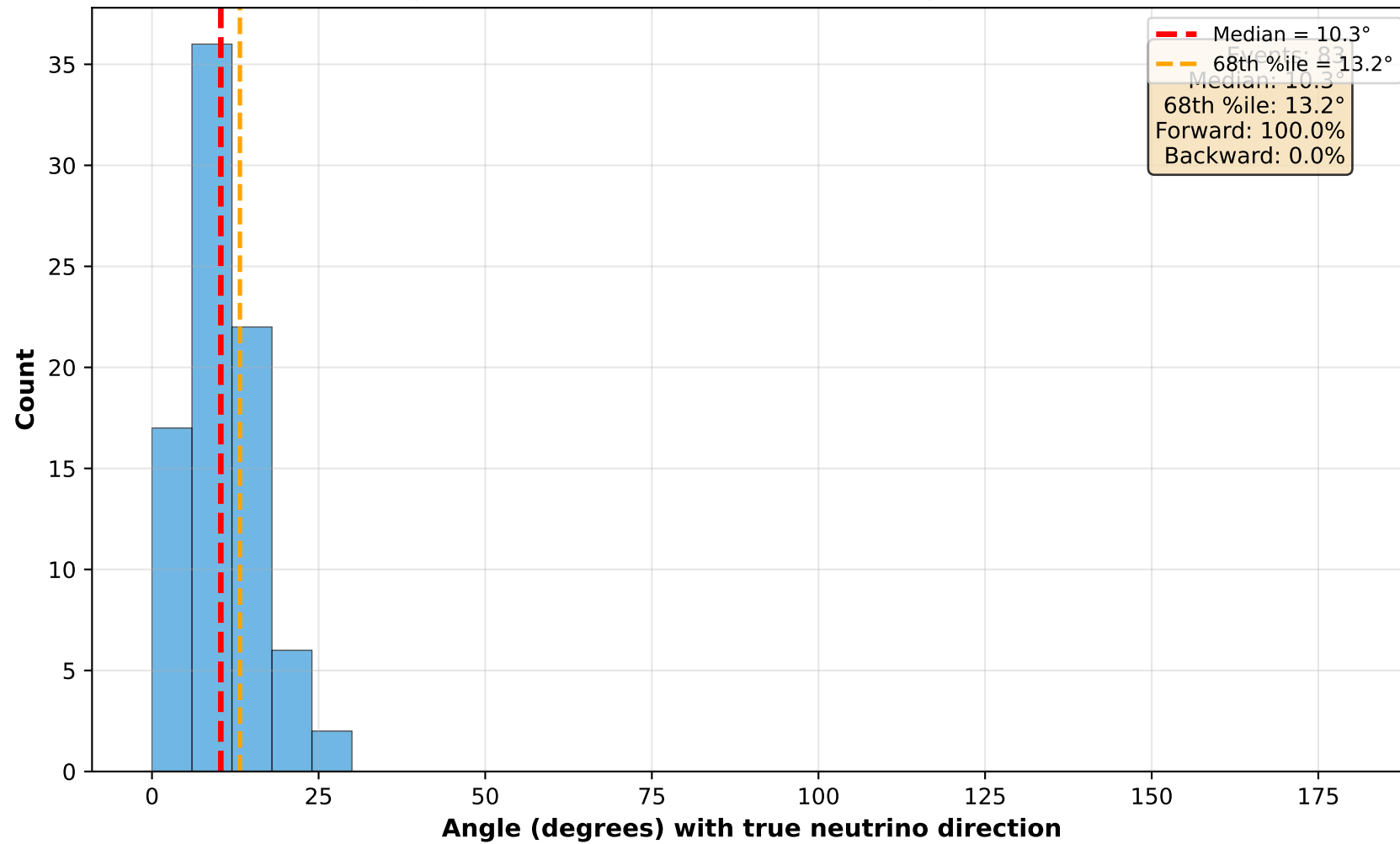
X-aligned (within $\pm 30.0^\circ$ of X-axis): Angle Distribution



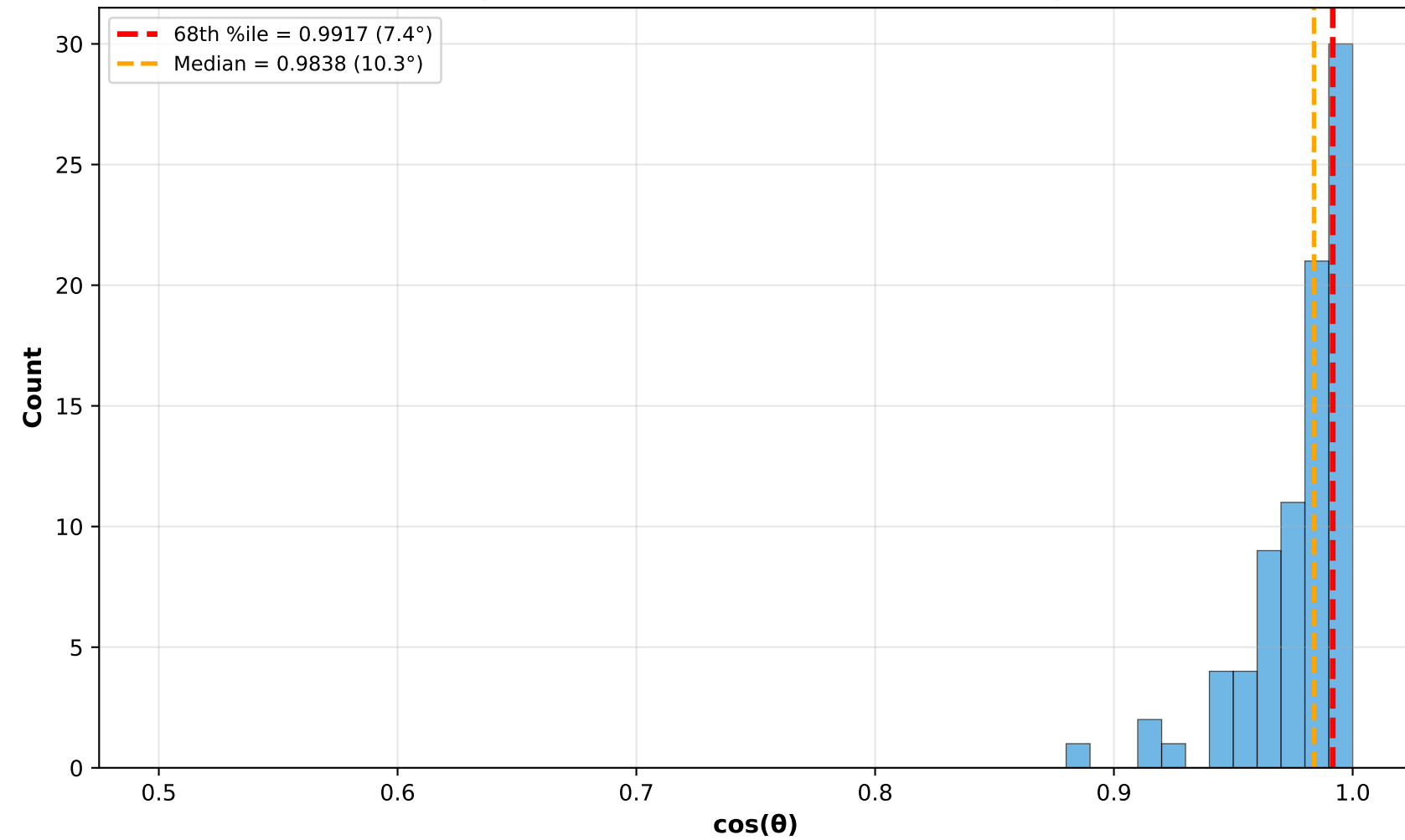
X-aligned: Cosine Distribution [0.5-1.0 range]



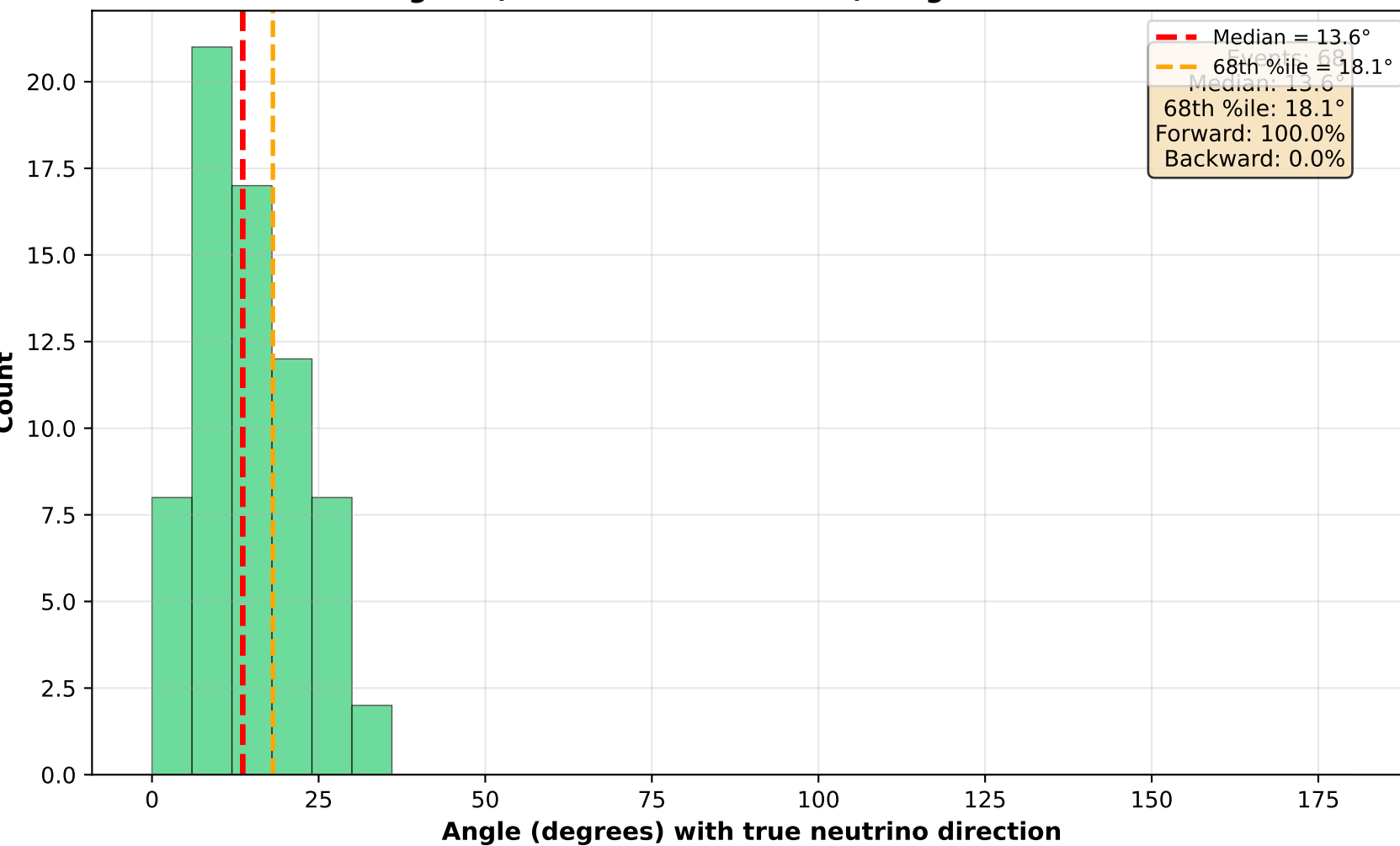
Y-aligned (within $\pm 30.0^\circ$ of Y-axis): Angle Distribution



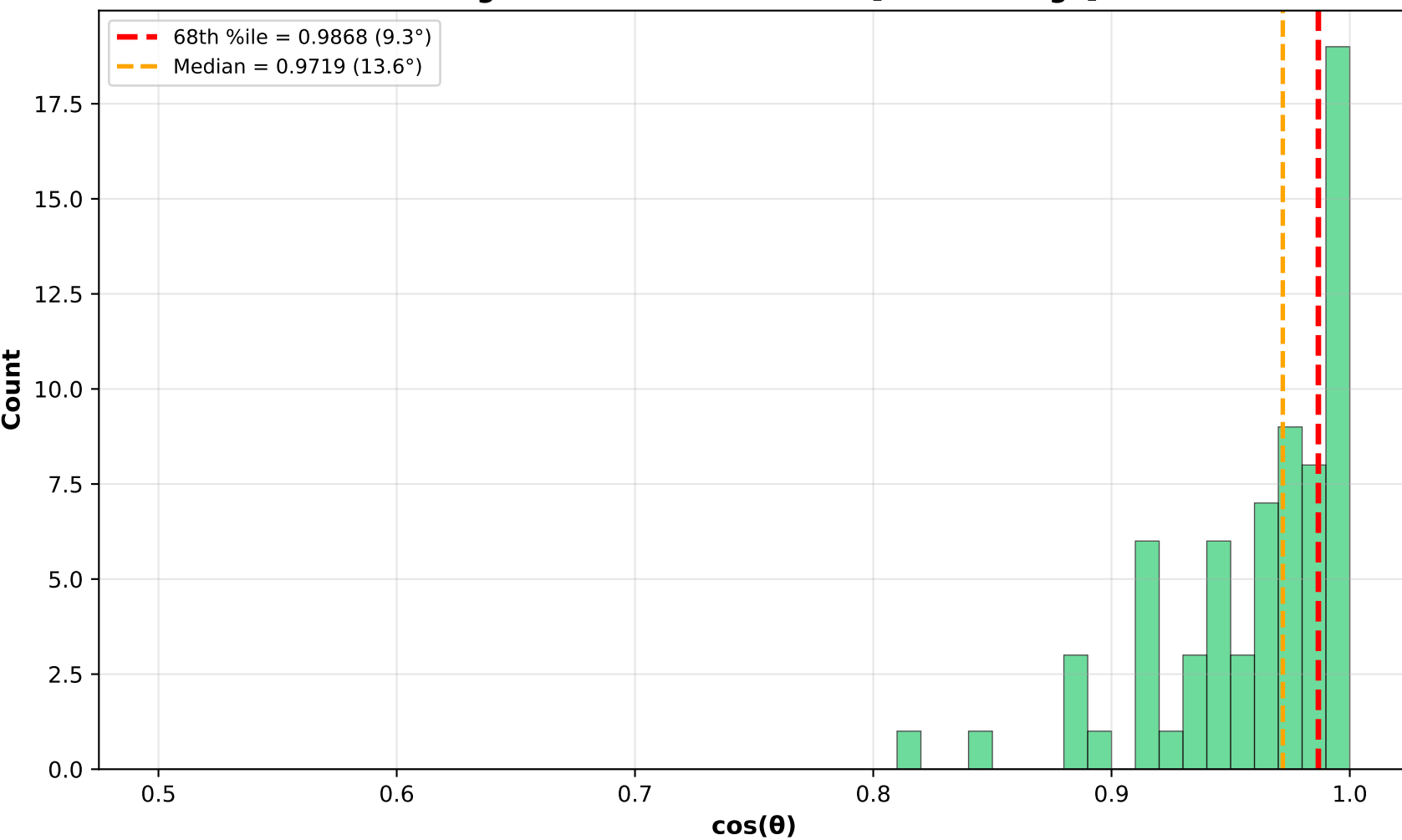
Y-aligned: Cosine Distribution [0.5-1.0 range]



Z-aligned (within $\pm 30.0^\circ$ of Z-axis): Angle Distribution

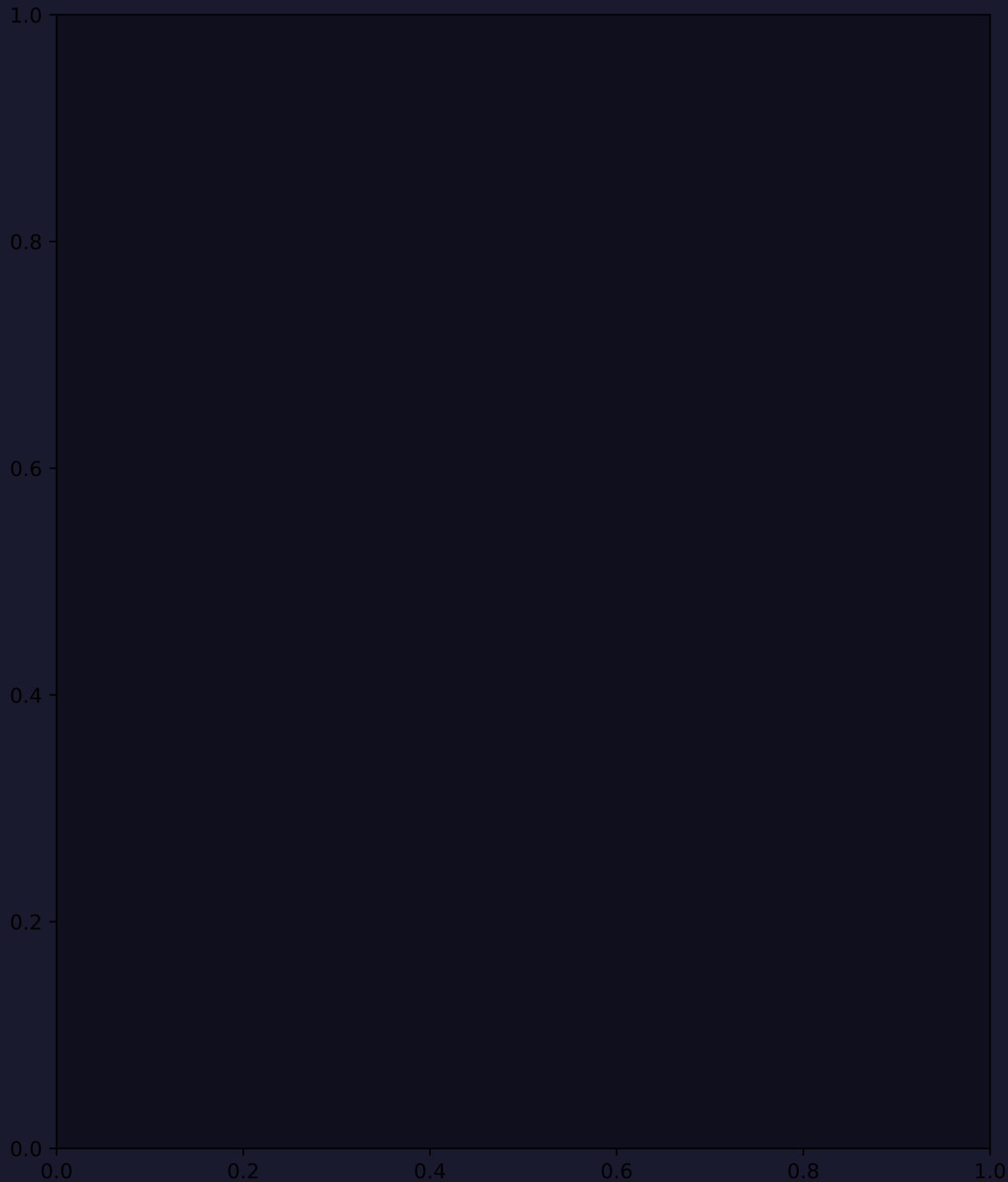
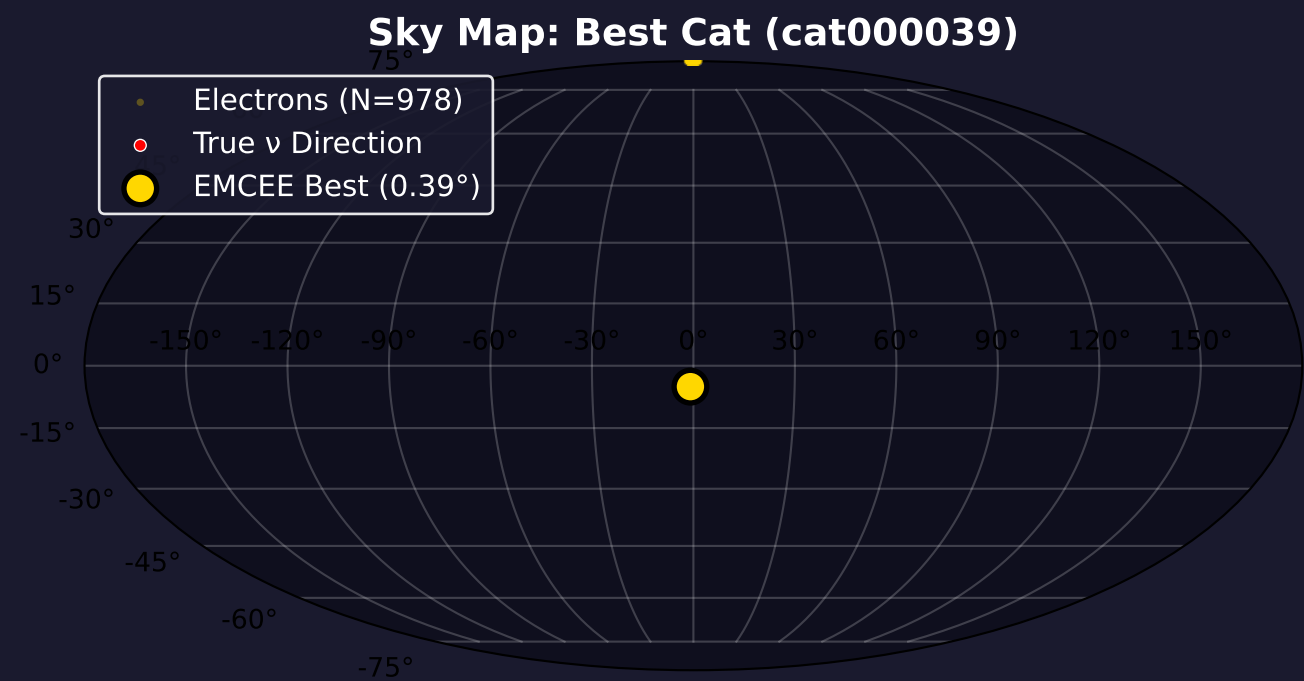


Z-aligned: Cosine Distribution [0.5-1.0 range]

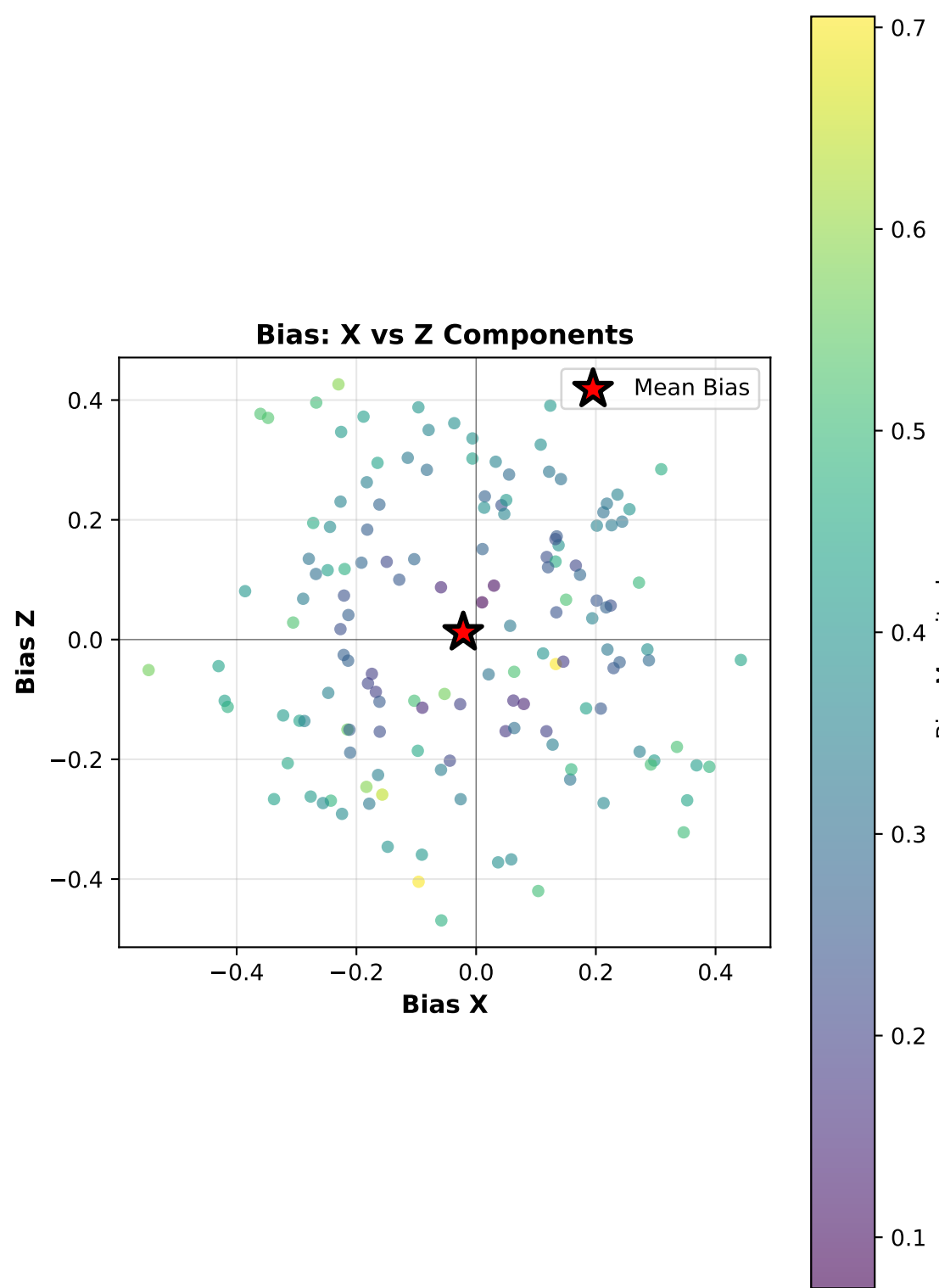
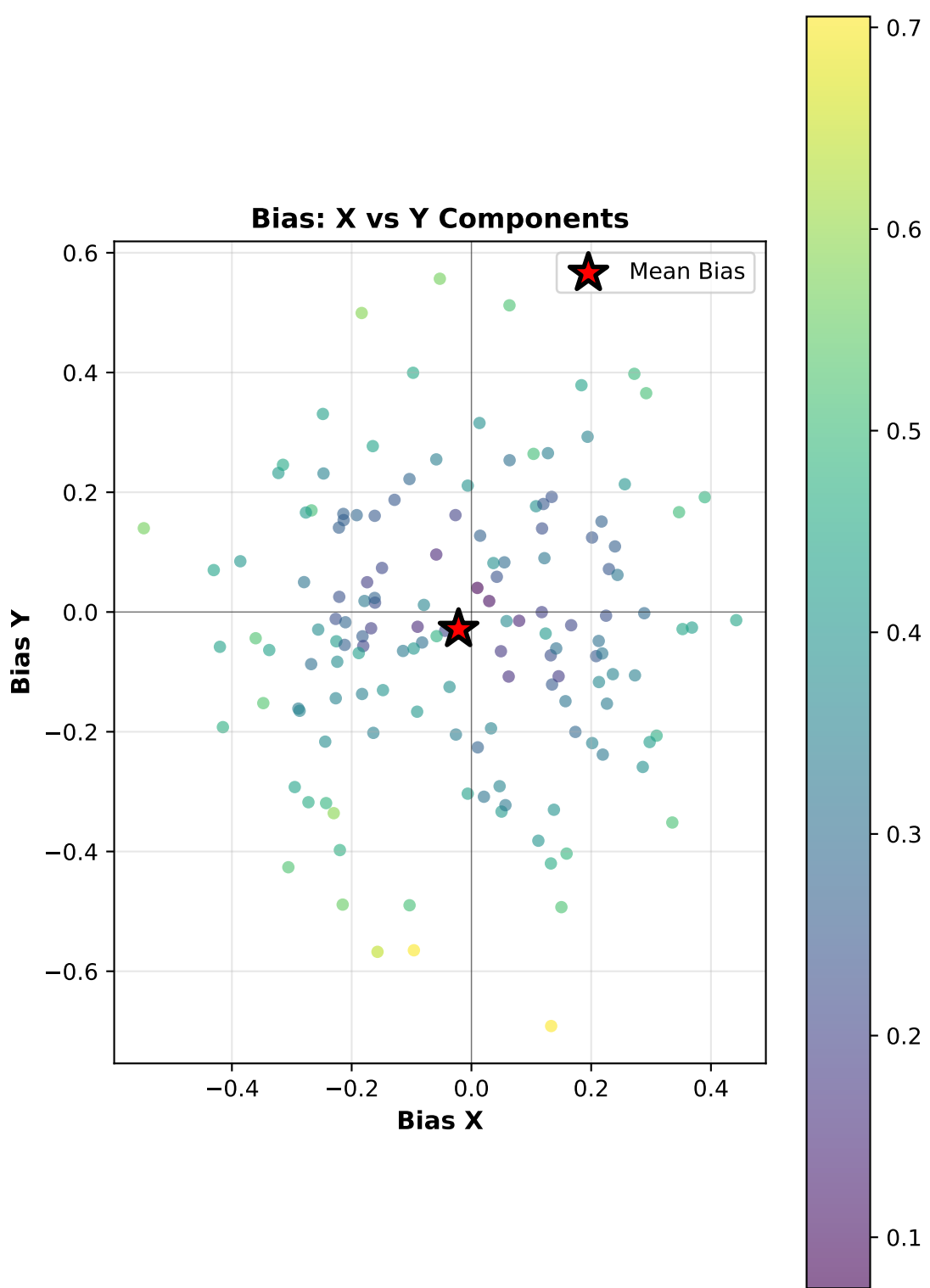
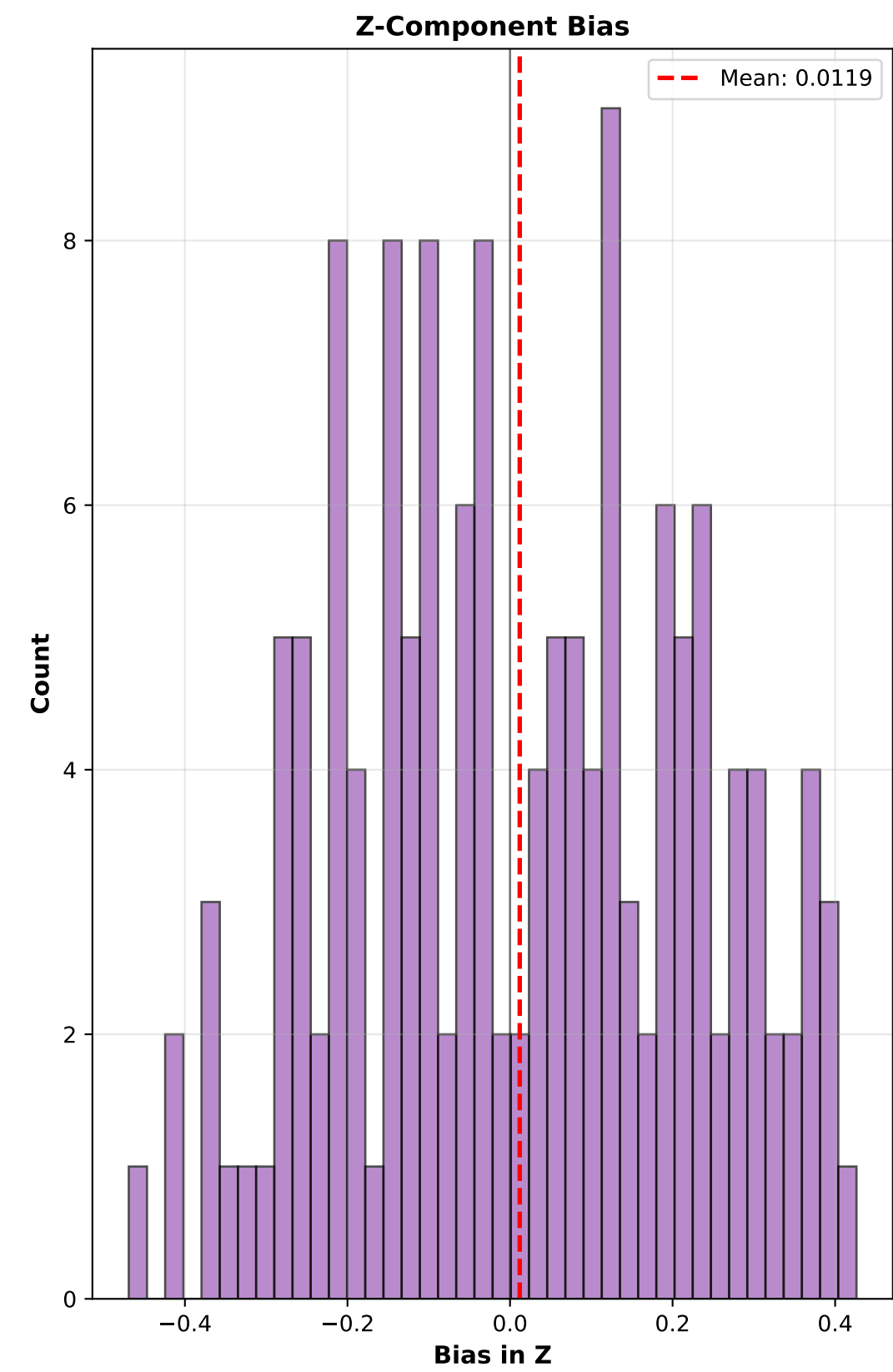
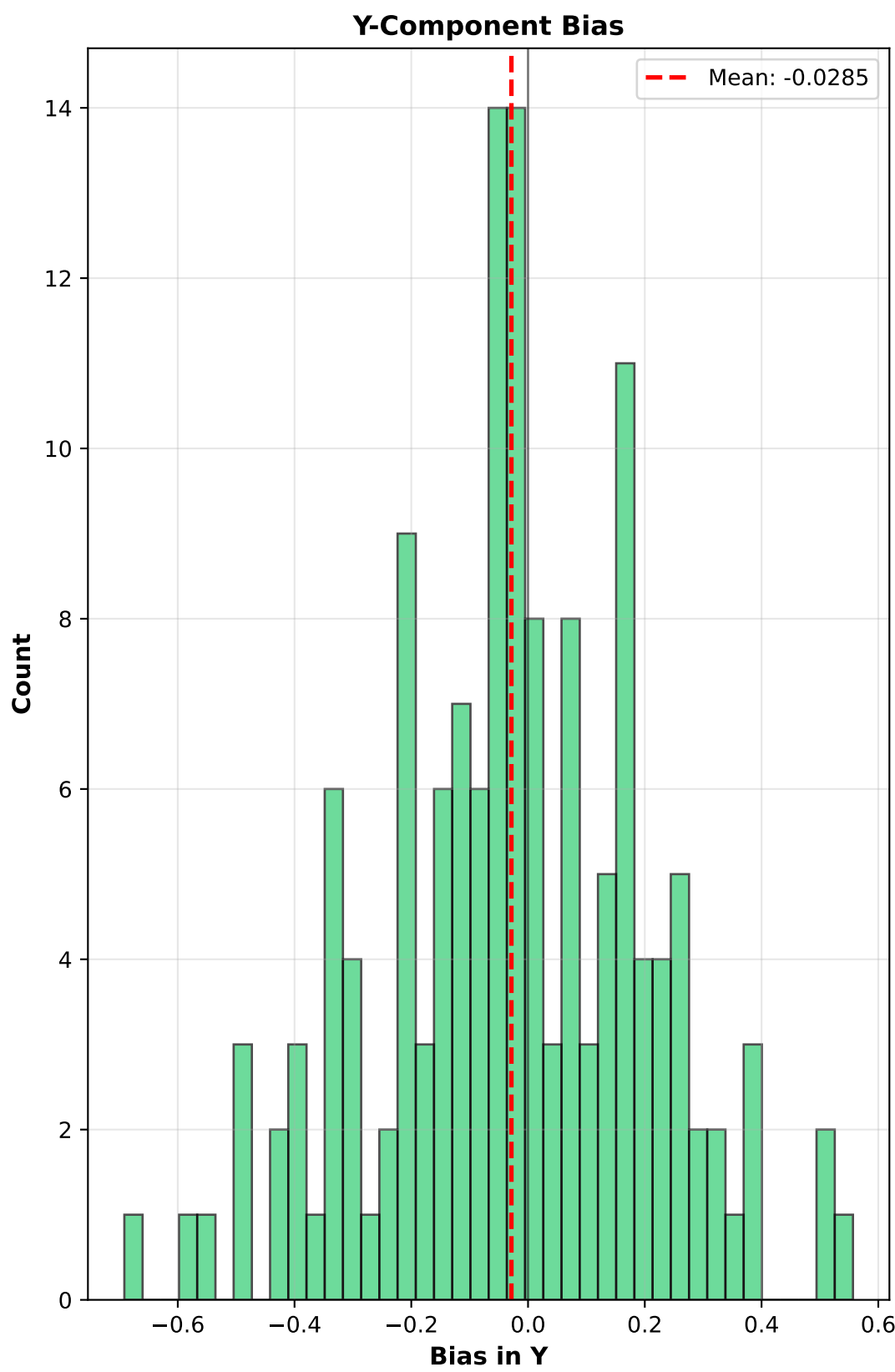
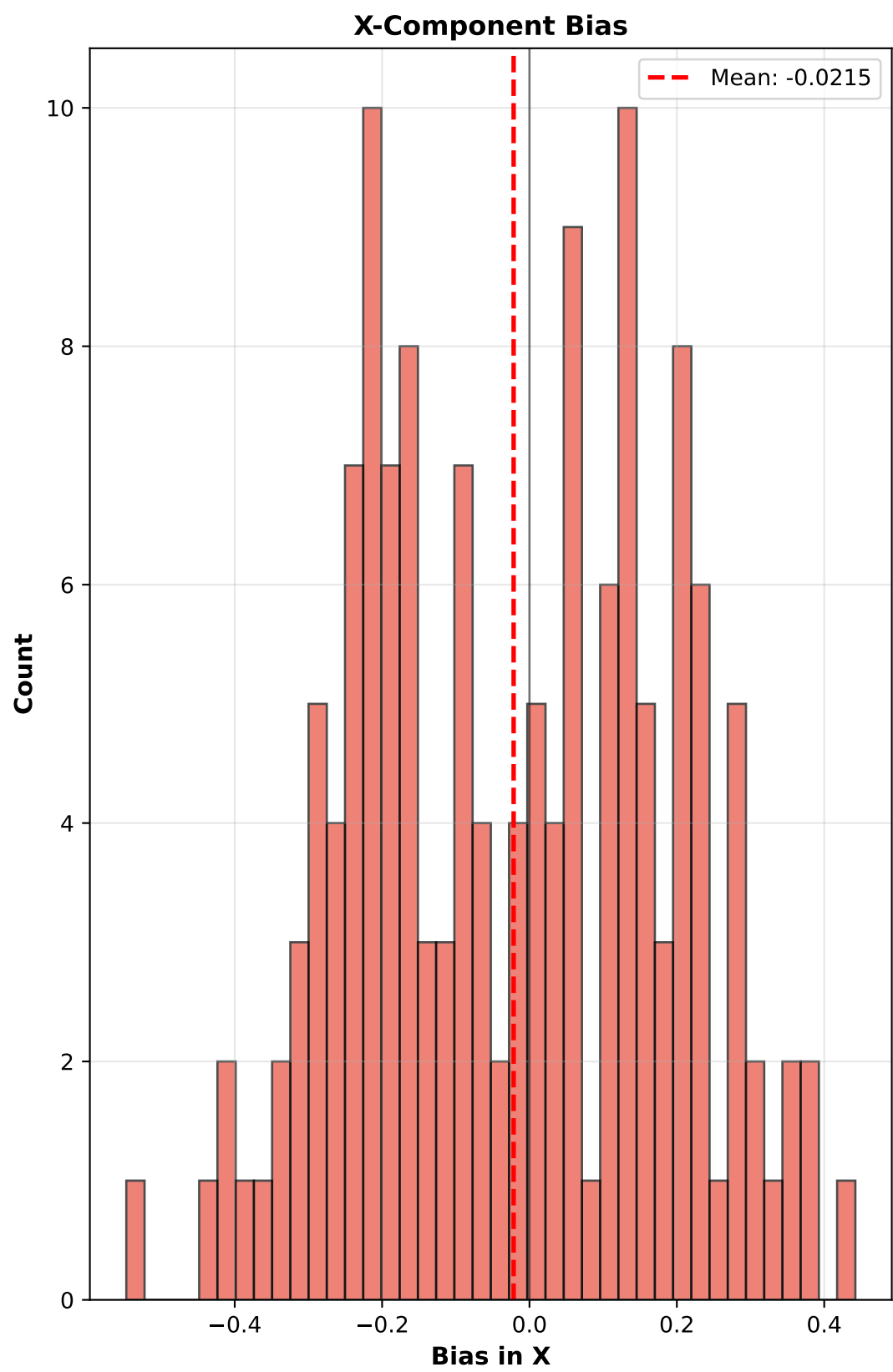
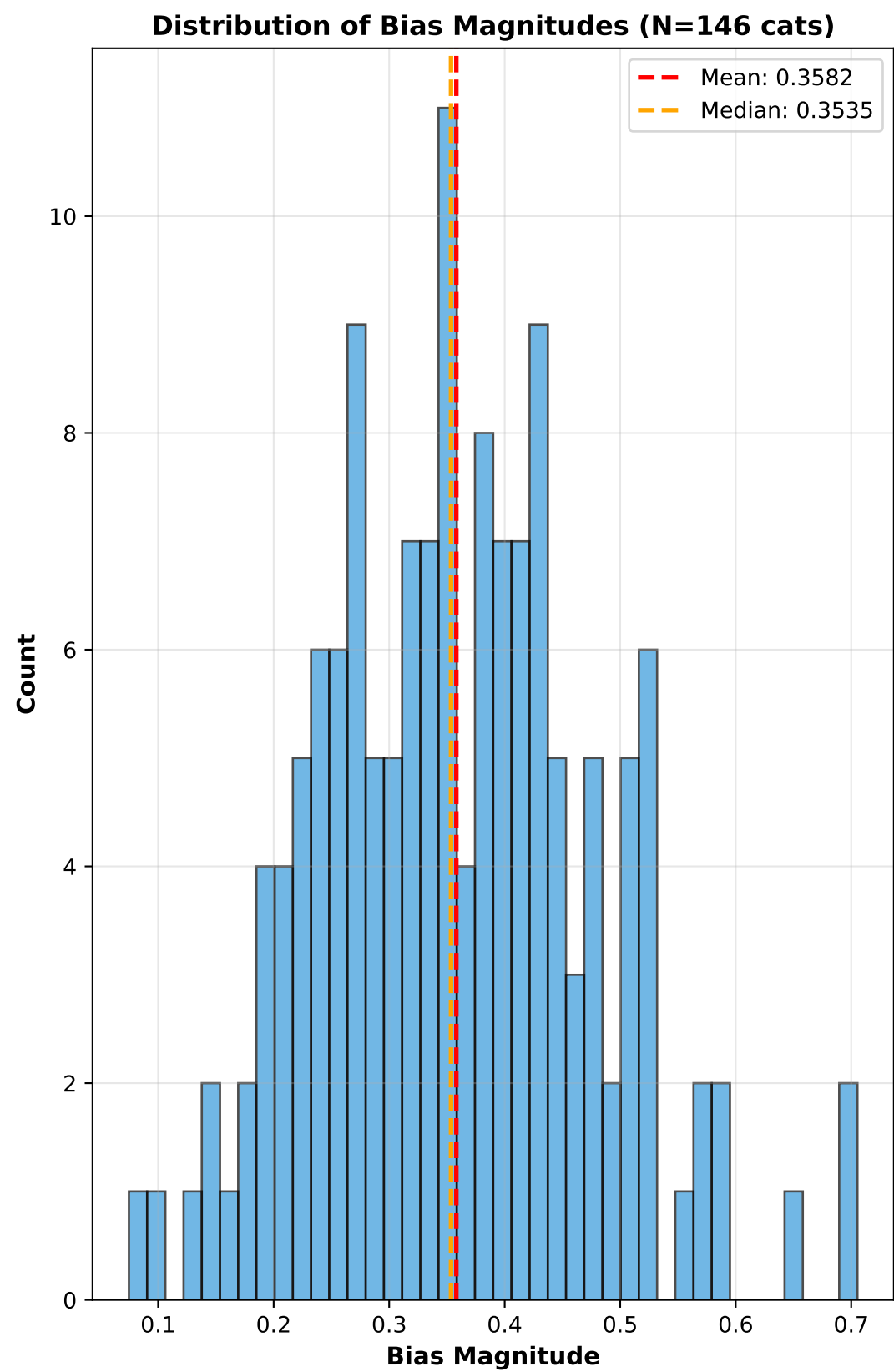


NOTE: Axis alignment is based on TRUE neutrino direction. X-axis performance depends on detector geometry (drift direction, wire orientation). Verify X/Y/Z coordinate definitions match your detector setup.

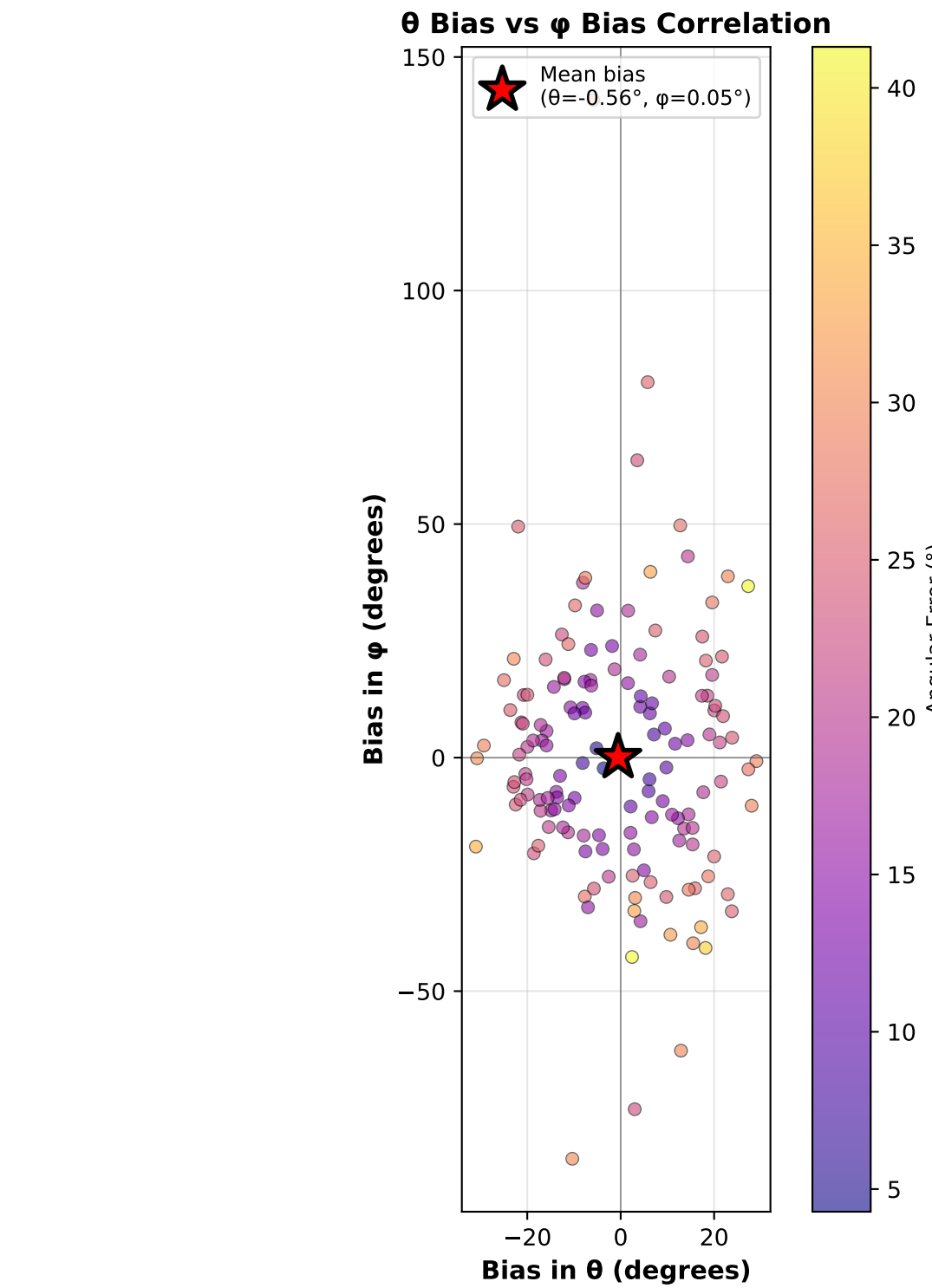
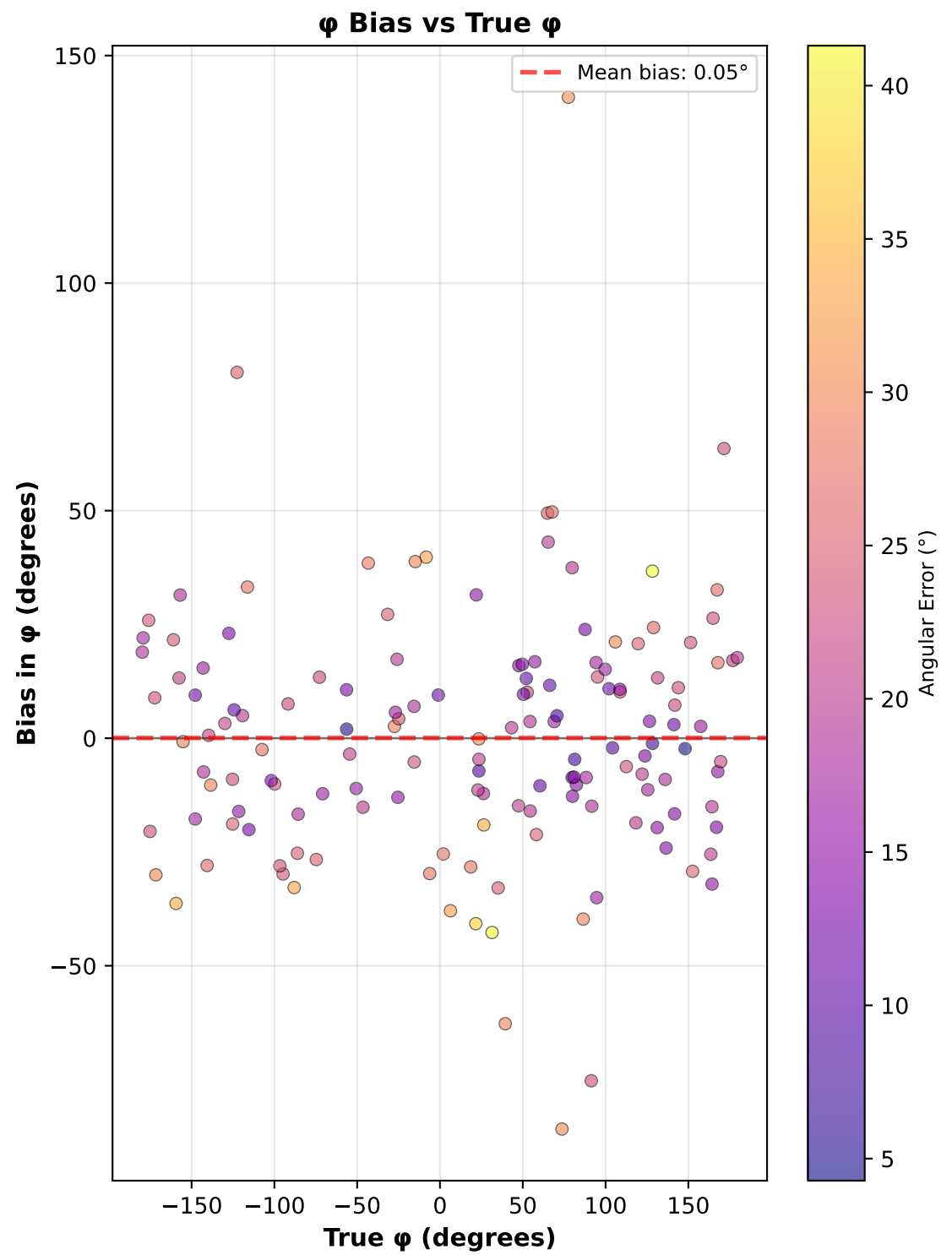
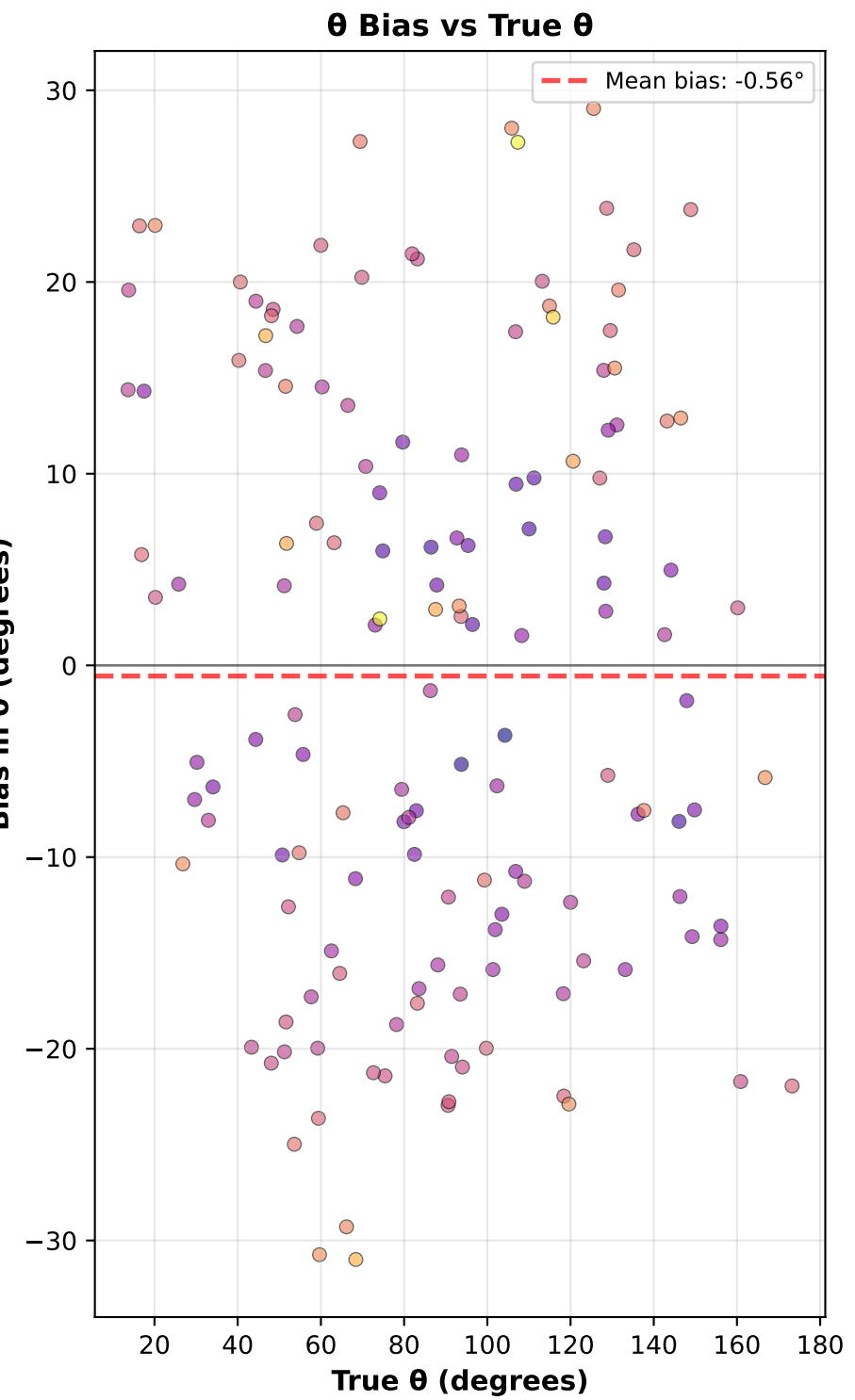
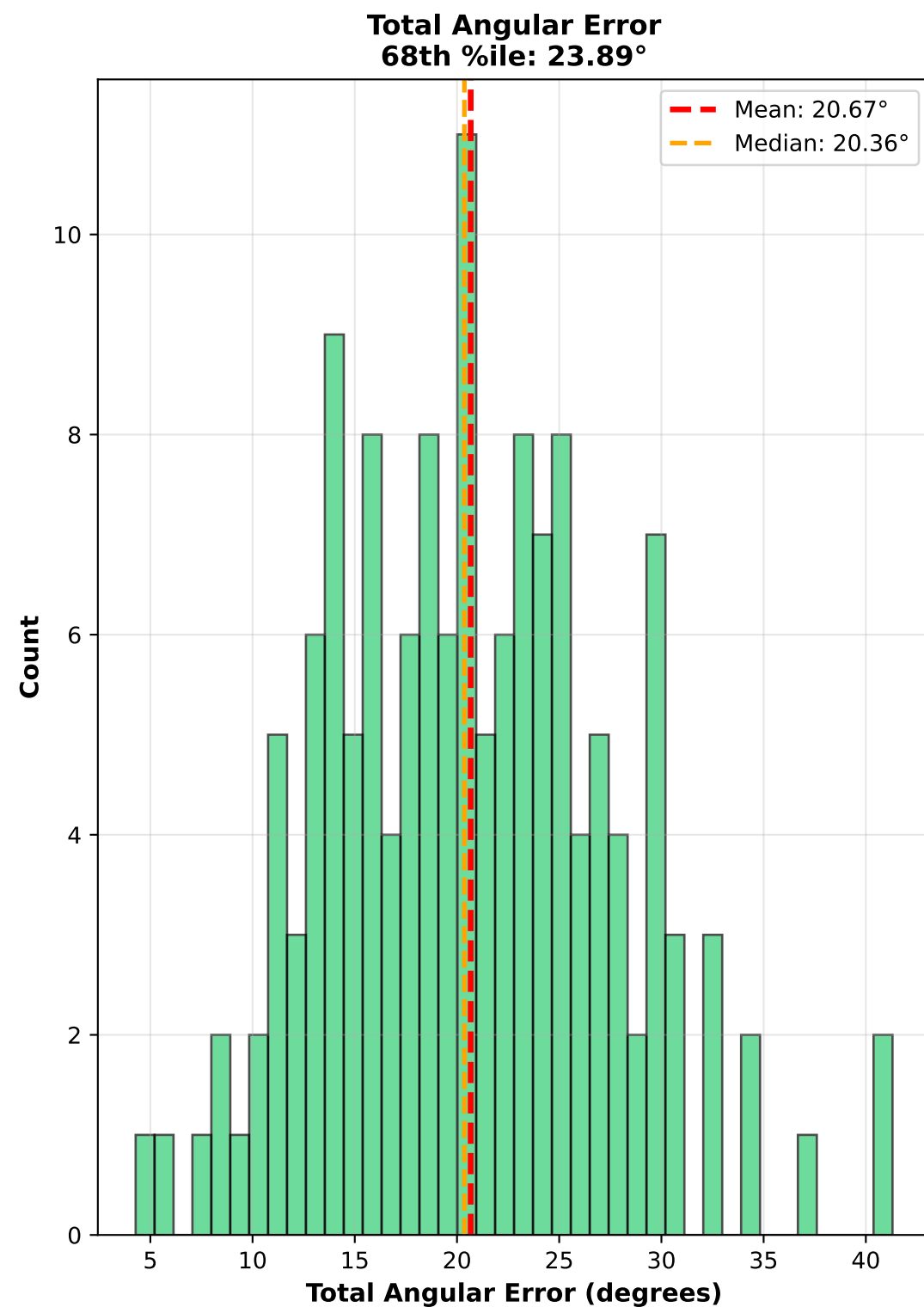
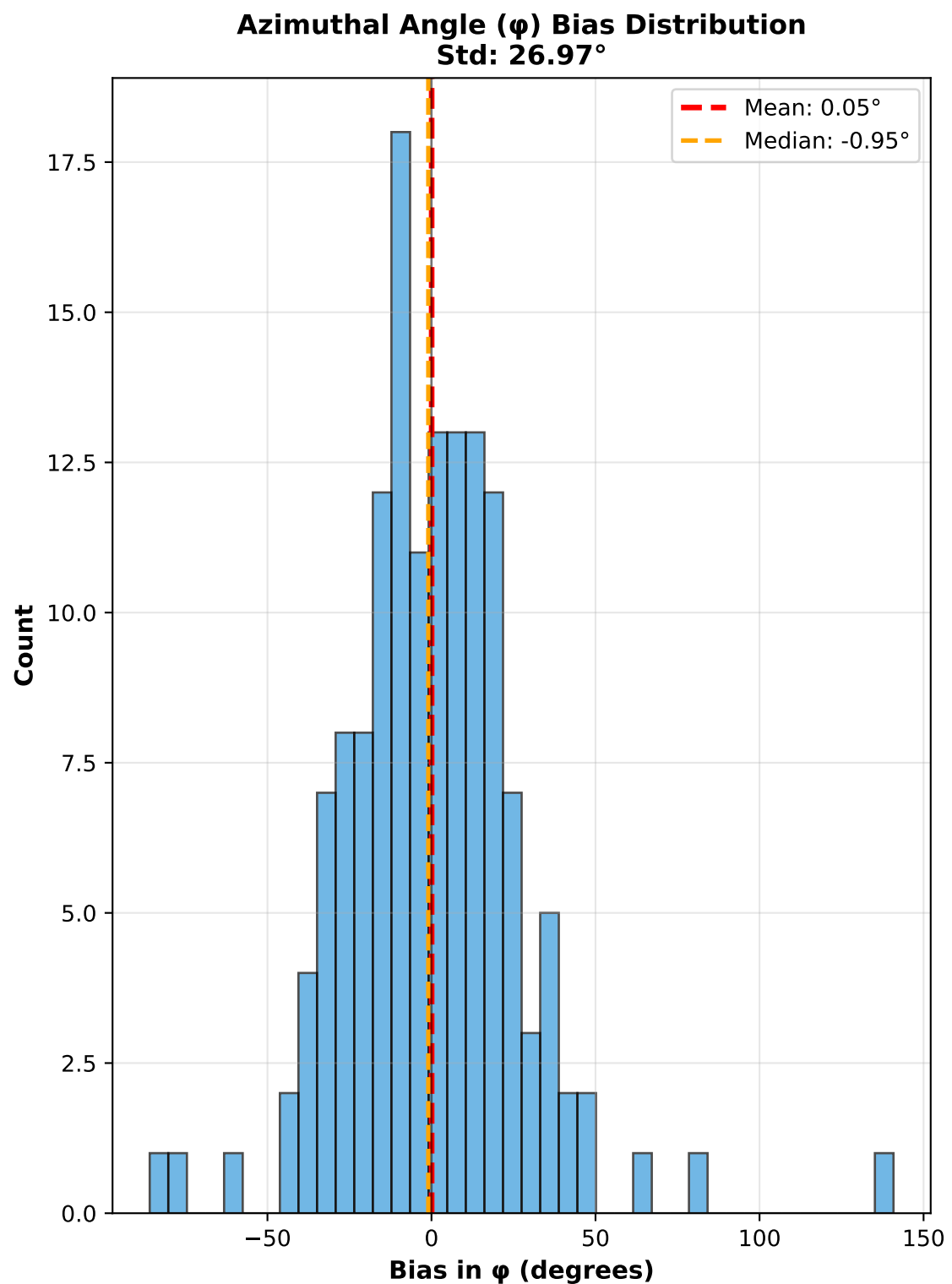
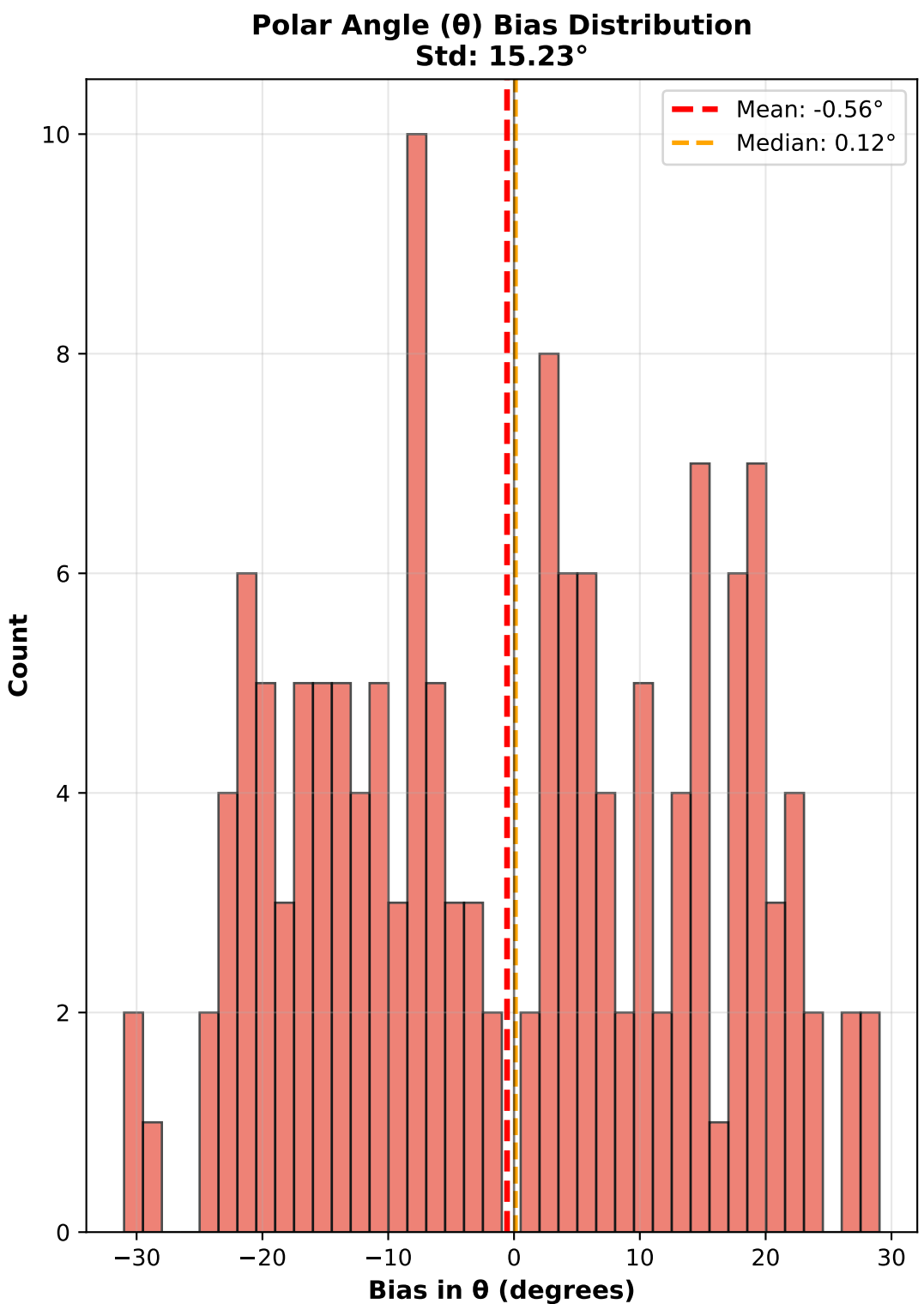
Neutrino Pointing Reconstruction: cat000039 - Perfect CT Scenario
Yellow: Electrons | Red: True Neutrino Direction



Directional Bias Analysis (Perfect CT) - Cartesian Components
Mean Bias Vector: (-0.0215, -0.0285, 0.0119), Magnitude: 0.0377



Angular Bias Analysis in Spherical Coordinates (Perfect CT)
 θ = polar angle from z-axis [0° , 180°], ϕ = azimuthal angle from x-axis [-180° , 180°]



Angular Bias Statistics (N=146 cats):
 θ bias: mean= -0.56° , median= 0.12° , std= 15.23°
 ϕ bias: mean= 0.05° , median= -0.95° , std= 26.97°
Total angular error: mean= 20.67° , 68th %ile= 23.89°